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**A Shariah-Governed MCDM Readiness Framework for Smart
Infrastructure Sukuk:**

Integrating Maqasid, ESG and Financial Viability
Research Proposal for the M.Sc in Islamic Finance

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1. Executive Abstract

Smart infrastructure sukuk need more than a formal compliance check. They require an appraisal logic that can read financial strength, Shariah structure, sustainability value and operational performance together, inside one auditable framework. Current screening practice still splits this work into separate lanes: financial analysts test cash flows and repayment capacity; sustainability teams review ESG disclosures; technical experts assess infrastructure performance; and Shariah boards examine contractual permissibility. That separation creates the core problem at the pre-structuring stage: projects may look eligible, yet remain hard to compare across Maqasid al-Shariah, ESG, financial, technical and risk criteria. This study develops, operationalizes and validates the Shariah-Compliant Smart Infrastructure Investment Readiness Score (SC-SIIRS), a transparent multi-criteria decision-making framework for screening smart infrastructure projects before sukuk structuring. The design is sequential and practical. AHP will derive expert weights from 8-12 specialists; TOPSIS will rank three to five representative scenarios; MEREC and COPRAS will test robustness; and 6-10 semi-structured interviews will examine Shariah-governance fit and practical usability. The contribution is a project-level decision-support architecture that organizes evidence, exposes trade-offs and supports expert judgment without replacing Shariah boards or investment committees.

2. Introduction and Background

Smart infrastructure sukuk now stand at the meeting point of three forces: the growth of Islamic finance, the pressure to fund sustainable infrastructure, and the demand for clearer investment appraisal tools. Islamic finance is no longer a peripheral market. The Islamic Financial Services Board reported that global Islamic financial services assets reached USD 3.88 trillion in 2024, with 14.9 percent year-on-year growth; sukuk grew even faster, at 25.6 percent (Islamic Financial Services Board, 2025).

Sukuk are Islamic investment certificates linked to underlying assets, usufructs or services. That asset-linked structure makes them highly relevant to infrastructure finance because capital is tied to real economic activity rather than to purely financial claims. Infrastructure, however, has also changed. Cities are no longer financing only roads, utilities and buildings. They are investing in smart mobility, digital public infrastructure, intelligent transport networks, climate-resilient utilities and data-enabled urban services. These projects are capital-intensive, technically complex and impossible to judge through financial indicators alone.

The sustainable sukuk market has started to respond. ICMA-IsDB-LSEG guidance explains how sukuk can be labelled as green, social or sustainability instruments and aligned with recognized sustainable-finance principles (ICMA, Islamic Development Bank, & LSEG, 2024). Labelling, however, does not solve the deeper appraisal problem. When several infrastructure projects appear eligible for sustainable or Shariah-compliant financing, Islamic banks, sukuk issuers and Shariah boards still need a transparent way to compare them across financial viability, Maqasid alignment, ESG performance, smart infrastructure outcomes and risk exposure.

This problem is not theoretical. The World Bank and Islamic Development Bank estimate that only 8.2 percent of sustainable syndicated loans raised between 2017 and 2024 were Shariah-compliant, despite the scale of Islamic banking assets and the natural fit between Islamic finance and sustainable real-asset investment (World Bank & Islamic Development Bank, 2025). That figure points to a practical under-deployment of Islamic finance in sustainable and climate-aligned channels. The question is not whether Islamic finance has ethical principles. It is whether those principles can be translated into tools that help allocate capital toward measurable development outcomes.

Current appraisal practice remains fragmented. Financial analysts review capital expenditure, operating expenditure, projected cash flows and repayment capacity. Sustainability teams evaluate ESG disclosures. Technical experts assess operational performance, digital reliability and infrastructure resilience. Shariah boards examine contractual permissibility, asset ownership, revenue sources and compliance with Islamic finance standards. Each review is necessary. The weakness is that these reviews often run in parallel instead of feeding into one pre-screening architecture.

This fragmentation creates two risks. First, projects can receive a sustainability label without strong evidence of measurable impact, which increases the risk of greenwashing. Second, projects can satisfy formal Shariah structuring requirements while showing weak substantive alignment with Maqasid al-Shariah, especially on public welfare, harm reduction, environmental stewardship and equitable access. I describe this second risk as Shariah-washing: contractual compliance without convincing ethical or developmental substance.

The regulatory context raises the stakes. AAOIFI's exposure draft of Shariah Standard No. 62 on sukuk shows that sukuk governance remains an active area of standard-setting and market debate, especially around asset ownership, legal transfer and the substance of Shariah-compliant structures. The draft must be treated carefully because it is not a finalized universal rule. Still, its existence confirms that sukuk structuring is becoming more demanding and more closely scrutinized (AAOIFI, 2024).

Gap statement: the literature and market guidance still lack an operational project-level framework that integrates financial viability, Maqasid al-Shariah alignment, ESG performance, smart infrastructure outcomes, Shariah governance and risk exposure into one auditable investment readiness score for smart infrastructure sukuk. Existing work addresses the pieces. Islamic finance scholarship explains Shariah principles and Maqasid objectives; sustainable finance frameworks define green or social eligibility; smart-city assessment models evaluate urban performance; and MCDM methods rank complex alternatives. What remains missing is their integration into a practical decision-support model for the pre-structuring stage of sukuk finance.

This study occupies that niche by proposing the Shariah-Compliant Smart Infrastructure Investment Readiness Score (SC-SIIRS). It will design, operationalize and validate a transparent multi-criteria decision-support framework for evaluating smart infrastructure projects before sukuk structuring. The model combines AHP-based expert

weighting, TOPSIS project ranking, MEREC and COPRAS robustness checks, and qualitative validation through semi-structured expert interviews. It does not replace Shariah boards, investment committees or legal structuring teams. Its role is narrower and more useful: to organize evidence, improve comparability and support pre-screening before formal sukuk approval.

The scope is deliberately limited. The study focuses on smart infrastructure projects that could plausibly be financed through Shariah-compliant sukuk, including smart mobility systems, intelligent transport infrastructure, digital utilities and climate-resilient urban assets. It does not price sukuk, issue a Shariah ruling, draft legal documentation, audit post-issuance impact or build an autonomous artificial intelligence system. Its contribution is an auditable project-level appraisal framework that brings financial analysis, Maqasid-based ethical review, ESG assessment, technical performance and governance risk into one structured model.

The study matters in two ways. Academically, it moves the discussion beyond broad claims that Islamic finance and sustainability are compatible by proposing a testable appraisal architecture. Practically, it gives Islamic banks, sukuk issuers, Shariah boards, regulators and infrastructure planners a structured tool for comparing candidate projects before scarce capital is committed. The contribution is not another descriptive account of Islamic finance growth. It is a decision-support framework for making Islamic sustainable infrastructure finance more transparent, comparable and defensible.

3. Research Problem

Islamic finance rests on Shariah compliance, asset linkage, the avoidance of *riba*, excessive *gharar* and *maysir*, and the broader ethical objectives of *Maqasid al-Shariah*. In principle, this makes Islamic finance well suited to infrastructure projects that create real economic activity and public benefit. In practice, however, smart infrastructure projects remain difficult to appraise for sukuk financing because the relevant evidence is scattered across separate expert domains.

The fragmentation appears across four parallel workstreams:

- financial analysts assess CAPEX, OPEX, expected cash flows, repayment capacity, payback period, NPV and investor distribution coverage;
- sustainability teams assess ESG labels, climate benefits, disclosure quality and reporting methodology;
- technical teams assess smart infrastructure feasibility, interoperability, cybersecurity, system reliability and operational performance;
- Shariah boards review contractual permissibility, asset ownership, revenue sources and compliance with applicable Shariah standards (AAOIFI, 2024; AAOIFI & IFSB, 2022).

All four assessments are necessary. The problem is institutional fragmentation: they are often performed in parallel rather than integrated into one evidence-based decision-support architecture. As a result, a project can look financially viable but weak in public welfare, green-labelled but poorly supported by measurable impact data, or formally Shariah-compliant while failing to demonstrate substantive *Maqasid* alignment.

The research problem is the lack of a transparent, project-level, Shariah-governed scoring framework that can integrate financial viability, *Maqasid* alignment, ESG contribution, smart infrastructure performance and governance risk into an auditable pre-screening tool for smart infrastructure sukuk.

4. Research Gap and Novelty Defense

4.1 What the literature already establishes

The adjacent literature establishes three points that this proposal can build on rather than repeat. First, sukuk are now a major Islamic capital-market instrument because their asset-linked structure connects capital to real assets, usufructs or services rather than to interest-bearing debt claims. This is not a speculative claim: the Islamic Financial Services Board reports that total Islamic financial services assets reached USD 3.88 trillion in 2024, while sukuk expanded by 25.6 percent year-on-year (Islamic Financial Services Board, 2025).

Second, sustainable sukuk have moved from a conceptual niche into a recognized market category. ICMA-IsDB-LSEG guidance clarifies how sukuk can be labelled as green, social or sustainability instruments and aligned with sustainable-finance principles (ICMA, Islamic Development Bank, & LSEG, 2024). This guidance improves market consistency, but it mainly addresses eligibility, labelling and disclosure. It does not tell decision-makers which eligible infrastructure project should be prioritized before sukuk structuring.

Third, Islamic finance scholarship increasingly connects Maqasid al-Shariah with public welfare, harm reduction, ethical investment and environmental responsibility (Ahmed, 2011; Chapra, 2008; Dusuki & Abdullah, 2007). This literature matters because it prevents Shariah compliance from being reduced to contractual form. Its weakness is operational: it explains why Islamic finance should pursue sustainability, but it gives less guidance on how institutions should compare specific smart infrastructure projects with measurable, auditable criteria.

4.2 What remains missing: the precise research gap

Despite progress in sukuk finance, sustainable Islamic finance, Maqasid-based ethics, smart-city assessment and MCDM, the reviewed literature has not produced an operational project-level framework that integrates financial viability, Maqasid alignment, ESG contribution, smart infrastructure performance, Shariah governance and risk exposure into one auditable investment readiness score for smart infrastructure sukuk.

This is the precise gap. Sukuk are not unstudied. ESG and Islamic finance are not unrelated. MCDM is not absent from finance or infrastructure research. Each field is already established. The missing element is their operational integration at the pre-structuring stage,

where Islamic banks, sukuk issuers, Shariah boards and infrastructure planners must compare candidate projects before committing legal, financial and institutional resources.

This absence matters because project trade-offs are rarely linear. One project may offer strong financial returns but weak Maqasid contribution. Another may perform well on environmental criteria but have uncertain revenue logic or limited asset-transferability for sukuk. A third may satisfy formal Shariah conditions while delivering weak measurable public-welfare benefits. Without an integrated appraisal framework, these trade-offs remain implicit, subjective and hard to audit.

The market relevance of this gap is reinforced by recent evidence. The World Bank and Islamic Development Bank estimate that only 8.2 percent of sustainable syndicated loans raised between 2017 and 2024 were Shariah-compliant, despite the scale of Islamic finance assets and the conceptual compatibility between Islamic finance and sustainable real-asset investment (World Bank & Islamic Development Bank, 2025). The issue is therefore not simply theoretical; Islamic finance remains under-deployed in sustainable and climate-aligned financing channels.

4.3 Why existing research is insufficient

Islamic banking performance studies usually evaluate institutions rather than projects. Their ratios, efficiency measures and performance indices help benchmark banks, but they do not answer the project-level question at the center of this proposal: whether a smart mobility corridor, digital utility or climate-resilient urban asset is ready for sukuk financing.

Sustainable sukuk frameworks improve market labelling, but they do not solve project prioritization. They can indicate whether a use of proceeds is broadly green, social or sustainable; they do not rank several eligible projects under competing financial, Maqasid, ESG and infrastructure-performance criteria.

Maqasid-ESG scholarship is often persuasive at the conceptual level but under-specified at the project level. It explains why Islamic finance should support public welfare, environmental stewardship and wealth protection, but it rarely converts these objectives into indicators that experts can score, challenge and validate.

Smart-city assessment models usually evaluate cities, municipal readiness or technological maturity. They rarely ask whether an individual smart infrastructure asset has

permissible revenue streams, sukuk-compatible asset logic, governance transparency or adequate distribution coverage.

MCDM and AI-assisted decision-support tools improve comparability, but they do not automatically create Shariah legitimacy. Weighting, normalization, ranking and threshold decisions remain human design choices. In Islamic finance, that point is critical: a model that cannot explain why one project outranks another creates governance risk rather than solving it.

Table 2 consolidates the literature gap by showing that each research stream contributes a necessary but incomplete layer; none provides a project-level, Shariah-governed readiness model for smart infrastructure sukuk.

Table 2. Critical Gap Synthesis Across Adjacent Research Streams

Literature stream	Established contribution	Remaining limitation	Implication for SC-SIIRS
Sukuk and Islamic infrastructure finance	Shows the relevance of asset-linked Islamic finance for real-economy infrastructure.	Does not provide a project-level readiness score for smart infrastructure assets.	Defines the financing context and the need for pre-structuring screening.
Green, social and sustainability sukuk	Clarifies eligibility, labelling and disclosure expectations.	Does not rank competing eligible projects or integrate Maqasid and technical performance.	Requires a prioritization layer beyond label eligibility.
Maqasid-ESG literature	Demonstrates conceptual alignment between Islamic ethics, public welfare and sustainability.	Often remains normative and insufficiently operational at project level.	Requires measurable, expert-validated Maqasid indicators.
Smart-city assessment	Provides indicators for mobility, digital infrastructure, resilience and operational performance.	Evaluates city readiness more often than sukuk financing readiness.	Supplies technical indicators but needs Islamic finance constraints.
MCDM and AI-assisted decision support	Offers transparent tools for weighting and ranking multi-criteria alternatives.	Rarely embeds Shariah governance, human oversight and auditability.	Provides the computational logic, but must remain advisory and Shariah-governed.

Note. SC-SIIRS = Shariah-Compliant Smart Infrastructure Investment Readiness Score; MCDM = multi-criteria decision-making. The table summarizes how each literature stream contributes to the proposal while leaving unresolved the project-level pre-structuring readiness problem.

4.4 How this research fills the gap

This study fills the gap by developing SC-SIIRS, an auditable decision-support framework for screening smart infrastructure projects before sukuk structuring. Its contribution rests on five linked design choices.

First, the unit of analysis is the project, not the Islamic bank, the city or the sukuk market as a whole. This matters because capital-allocation decisions occur before issuance, when decision-makers must assess whether a candidate asset deserves the cost of legal structuring, Shariah review and investor preparation.

Second, the model integrates evaluation dimensions that are usually assessed separately: financial viability, Maqasid al-Shariah alignment, ESG contribution, smart infrastructure performance, Shariah governance and risk exposure.

Third, the framework combines expert judgment and computational ranking. AHP is used for expert-derived weighting, TOPSIS for primary project ranking, and MEREC/COPRAS for robustness checks. This hierarchy prevents method inflation: AHP and TOPSIS are the core model, while MEREC and COPRAS test sensitivity rather than adding unnecessary complexity.

Fourth, the framework preserves Shariah governance. SC-SIIRS does not issue a fatwa, certify Shariah compliance or replace the Shariah board. It organizes evidence and makes trade-offs visible before formal approval.

Fifth, the framework responds to a current market need. The expansion of sukuk and sustainable finance increases the need to distinguish projects that are merely eligible from projects that are genuinely investment-ready.

4.5 Novelty defense

The originality of this study does not come from inventing Maqasid, ESG, sukuk, smart-city assessment or MCDM. Those fields already exist. The novelty lies in integrating them into a project-level, pre-structuring, Shariah-governed scoring architecture for smart infrastructure sukuk.

This distinction is essential. A proposal that simply states that Islamic finance supports sustainability would be derivative. A comparison of Islamic and conventional banks would be overdone. A general discussion of green sukuk would remain too broad. This study

is narrower and more defensible: it asks how Islamic financial institutions can evaluate and compare smart infrastructure projects before sukuk structuring, using transparent, auditable and multi-dimensional criteria.

The contribution is both theoretical and practical. Theoretically, the study moves Islamic sustainable finance from conceptual alignment toward operational decision-support. Methodologically, it adapts MCDM tools to a Shariah-sensitive appraisal problem. Practically, it offers sukuk issuers, Islamic banks, Shariah boards, regulators and infrastructure planners a structured pre-screening framework before capital and legal resources are committed.

The gap and the proposed solution therefore fit directly. The gap is the absence of an integrated project-level readiness model. The solution is SC-SIIRS: a transparent, MCDM-based and Shariah-governed investment readiness score for smart infrastructure sukuk.

5. Research Aim, Objectives and Questions

5.1 Primary Aim

The primary aim of this study is to develop, operationalize and validate the Shariah-Compliant Smart Infrastructure Investment Readiness Score (SC-SIIRS), a transparent MCDM framework for the ex-ante pre-screening of smart infrastructure projects considered for Shariah-compliant sukuk financing.

5.2 Specific Objectives

The objectives are deliberately limited to four sequential tasks. Each objective produces a concrete output and maps directly onto one research question and one methodological component.

1. To operationalize the SC-SIIRS criteria hierarchy by defining measurable indicators, data sources and scoring rules across financial viability, Maqasid al-Shariah alignment, ESG contribution, smart infrastructure performance, Shariah governance and risk exposure.

2. To derive and test criterion weights through an Analytic Hierarchy Process (AHP) expert survey with a minimum viable sample of 8-12 experts, using consistency screening and documented aggregation rules.

3. To rank three to five representative smart infrastructure sukuk scenarios using TOPSIS as the primary ranking method and to assess ranking stability through MEREC and COPRAS robustness checks.

4. To evaluate the Shariah-governance acceptability and practical usability of SC-SIIRS through six to ten semi-structured expert interviews, and to translate the findings into responsible pre-screening guidance for Islamic finance decision-makers.

5.3 Research Questions

Main research question: To what extent can a transparent MCDM-based readiness framework support the ex-ante pre-screening of smart infrastructure projects for Shariah-compliant sukuk financing without replacing formal Shariah review or investment committee judgment?

RQ1. Which measurable indicators, data sources and scoring rules are required to operationalize SC-SIIRS across financial, Maqasid, ESG, infrastructure, Shariah-governance and risk dimensions?

RQ2. What relative weights do qualified experts assign to the SC-SIIRS dimensions and sub-criteria, and how consistent are those judgments under AHP consistency screening?

RQ3. How do candidate smart infrastructure sukuk scenarios rank under TOPSIS, and how stable are those rankings under MEREC and COPRAS robustness checks?

RQ4. What governance, explainability and institutional conditions do experts identify as necessary for the responsible use of SC-SIIRS in sukuk pre-screening?

5.4 Alignment and Scope Control

The section uses a mirror structure: Objective 1 answers RQ1 through criteria operationalization; Objective 2 answers RQ2 through AHP weighting; Objective 3 answers RQ3 through project ranking and robustness testing; and Objective 4 answers RQ4 through expert validation. This structure controls the scope. The study does not price sukuk, issue a fatwa, conduct legal documentation, audit post-issuance impact or build an autonomous AI system. Its output is a pre-screening decision-support framework, not a substitute for Shariah boards or investment committees.

6. Critical Literature Review

This review is organized around analytical fault lines rather than author-by-author chronology. It asks how five research streams - sukuk infrastructure finance, sustainable sukuk, Maqasid-ESG scholarship, smart-city assessment and MCDM-based decision support - explain the proposed study, and where each stream fails to answer the project-level question at its center. The review therefore does not ask whether these literatures exist. It asks why their existing evidence remains insufficient for pre-screening smart infrastructure projects for Shariah-compliant sukuk financing.

6.1 Sukuk infrastructure finance: structural compatibility does not equal project readiness

The first consensus is clear: sukuk are structurally relevant to infrastructure because they connect financing to underlying assets, usufructs or services rather than to conventional interest-bearing claims. Institutional guidance on Islamic infrastructure finance and recent market reporting support this consensus by presenting sukuk as a credible instrument for real-asset financing and public-private investment (World Bank & Islamic Development Bank, 2019; Islamic Financial Services Board, 2025). This literature is valuable because it establishes the macro-financial legitimacy of sukuk as an infrastructure instrument.

However, the same literature is weaker at the micro level where investment decisions are actually made. Most studies and reports address sukuk structures, market development, legal compatibility or sectoral potential; they rarely specify how a particular smart mobility corridor, digital utility platform or climate-resilient infrastructure asset should be compared against another before sukuk structuring begins. This is a serious limitation because a project can be asset-compatible yet still be weak in repayment capacity, Maqasid contribution, climate resilience, data governance, technical interoperability or Shariah auditability. The limitation is therefore not a lack of discussion on sukuk and infrastructure, but a lack of project-level investment readiness logic.

6.2 Sustainable sukuk: eligibility frameworks are not prioritization models

The second research stream concerns sustainable sukuk. Recent institutional guidance has improved the market language around green, social and sustainability sukuk by clarifying how issuers may label sukuk and align them with recognized sustainable finance principles (ICMA, Islamic Development Bank, & LSEG, 2024). Similarly, the IsDB Sustainable

Finance Framework links eligible financing to project categories, assets, capital expenditure, operating expenditure and impact reporting indicators (Islamic Development Bank, 2025). These frameworks are important because they reduce ambiguity around use-of-proceeds classification and post-issuance impact reporting.

Their weakness is that eligibility is not prioritization. A framework can indicate whether a project is broadly green or social, but it does not necessarily rank several eligible projects when capital is limited. It also does not resolve trade-offs between high capital expenditure and high environmental benefit, strong public welfare and weak cash-flow visibility, or attractive repayment coverage and limited Maqasid impact. This matters because the World Bank and Islamic Development Bank identify a persistent under-deployment of Shariah-compliant finance in sustainable financing channels, including the small share of Shariah-compliant sustainable syndicated loans between 2017 and 2024 (World Bank & Islamic Development Bank, 2025). The practical bottleneck is therefore not simply market eligibility; it is the absence of a comparative appraisal tool that can distinguish eligibility from readiness.

6.3 Maqasid and ESG: conceptual convergence without sufficient project-level measurement

A third stream links Maqasid al-Shariah with sustainable and responsible finance. Foundational works argue that Islamic finance should not stop at formal contractual compliance. Chapra (2008) frames Maqasid as a development-oriented vision of human well-being; Dusuki and Abdullah (2007) connect Maqasid and *maslahah* to stakeholder-oriented responsibility; and Ahmed (2011) distinguishes between products that are substantively aligned with Maqasid and products that are merely compliant in form. Together, these works establish an important normative critique: Shariah compliance should be evaluated not only by what a transaction avoids, but also by what social and economic purposes it serves.

Recent ESG-Maqasid research extends this argument by developing disclosure frameworks, responsible-investment indices and sector-specific sustainability blueprints (Ramli et al., 2024; Zain et al., 2024). These contributions move the literature closer to measurement, but their dominant unit of analysis remains the institution, portfolio, index or sector. They are less effective for appraising a single infrastructure project before a sukuk transaction is structured. This creates a methodological gap for the present study: Maqasid indicators cannot be imported as abstract ethical language. They must be translated into

transparent proxies - such as public safety, accessibility, harm reduction, wealth preservation and environmental stewardship - and then subjected to expert validation because their weights and boundaries remain contestable.

6.4 Smart-city assessment: mature technical indicators, weak financing and Shariah sensitivity

Smart-city research has developed mature indicators for digital infrastructure, urban intelligence, mobility, public-service efficiency, resilience and data-enabled governance. MCDM-based city rankings demonstrate that digital and urban performance can be operationalized and compared across alternatives; Ye et al. (2022), for example, show how smart-city performance can be ranked using multi-criteria models. This stream is useful because it provides measurement logic for technical and operational dimensions that are relevant to smart infrastructure projects.

Yet the smart-city literature typically evaluates cities, municipal readiness or technology adoption rather than Islamic financing readiness. It rarely asks whether the underlying asset is suitable for sukuk, whether its revenue stream is permissible, whether ownership or usufruct can be transferred, whether the cash-flow profile can support sukuk distributions, or whether Shariah governance actors can audit the evidence. In other words, smart-city assessment is technically sophisticated but financially and jurisprudentially incomplete for the purpose of Shariah-compliant infrastructure financing. This gap justifies the inclusion of smart infrastructure performance in SC-SIIRS, but only as one dimension alongside financial viability, Maqasid alignment, Shariah governance and risk exposure.

6.5 MCDM and AI-assisted decision support: comparability without autonomous judgment

The fifth stream concerns multi-criteria decision-making. MCDM is appropriate for this proposal because the study problem contains conflicting criteria rather than one optimization target. AHP is suited to eliciting expert priorities where criteria involve normative judgment; TOPSIS ranks alternatives according to distance from ideal and anti-ideal solutions; MEREC tests objective criterion importance through removal effects; and COPRAS provides an alternative proportional ranking logic (Hwang & Yoon, 1981; Keshavarz-Ghorabae et al., 2021; Saaty, 1987; Zavadskas et al., 1994). These methods provide the technical basis for a transparent readiness score.

The critical weakness is that MCDM does not automatically produce legitimacy. Weights, normalization rules, benefit-cost classifications and threshold choices can alter rankings. In a Shariah-sensitive context, this is not a merely technical issue; it is a governance issue. A model that cannot explain why one project outranks another would be unsuitable for Shariah boards and investment committees. Therefore, this study treats MCDM as decision support, not autonomous judgment. AHP and TOPSIS form the core analytical engine, while MEREC and COPRAS function as robustness checks rather than decorative methodological additions.

6.6 Synthesis: from fragmented literatures to the SC-SIIRS research gap

The reviewed literature shows that the problem is not a lack of knowledge. It is a lack of integration. Sukuk scholarship explains asset-linked financing but rarely operationalizes project-level readiness. Sustainable sukuk guidance clarifies labels and eligible use of proceeds but does not rank competing eligible projects. Maqasid-ESG scholarship offers ethical direction but remains concentrated in institutional, portfolio or disclosure contexts. Smart-city assessment measures digital and urban performance but usually ignores sukuk structure and Shariah governance. MCDM methods enable ranking, but they require governance safeguards when applied to Islamic finance.

The resulting gap is precise: the reviewed literature has not produced an operational project-level framework that integrates financial viability, Maqasid al-Shariah alignment, ESG contribution, smart infrastructure performance, Shariah governance and risk exposure into one auditable investment readiness score for smart infrastructure sukuk candidates. This gap directly justifies the proposed SC-SIIRS framework. It also explains the methodology: AHP captures expert judgment on normative and technical priorities, TOPSIS produces the primary project ranking, MEREC and COPRAS test robustness, and expert interviews evaluate Shariah-governance acceptability and institutional usability.

The literature review therefore leads directly to the research design. The study does not claim to invent sukuk, Maqasid, ESG, smart-city assessment or MCDM. Its contribution is to connect them at the right unit of analysis - the candidate smart infrastructure project - and at the right decision moment: the pre-structuring stage of sukuk appraisal.

7. Theoretical Framework

7.1 Theoretical Positioning: Theory-Informed Framework Building

This study does not use theory as decoration. The theoretical framework defines what SC-SIIRS is allowed to measure, why those dimensions matter and how the results should be interpreted. The study is best understood as theory-informed framework building. It does not test a single universal theory of Islamic finance, and it does not claim to build theory from zero. It integrates established lenses to design a project-level decision-support framework for one decision problem: ex-ante screening before sukuk structuring.

The framework therefore uses theory in three ways. First, it defines investment worthiness beyond narrow financial return. Second, it explains why the current appraisal problem exists: financial, ESG, technical and Shariah evaluations are usually performed in separate institutional silos. Third, it translates those explanations into operational variables that can be measured, weighted, ranked and qualitatively validated.

7.2 Core Theoretical Logic

The central theoretical claim is simple: smart infrastructure sukuk readiness is not a one-dimensional financial construct. It is a composite governance construct shaped by ethical legitimacy, stakeholder welfare, sustainability performance, financial intermediation and decision-support transparency. Maqasid al-Shariah supplies the normative foundation; stakeholder theory extends the analysis beyond financiers and issuers; sustainable finance theory requires financial, social and environmental value to be assessed together; financial intermediation theory explains why structured evidence reduces information asymmetry; and decision-support systems theory explains why decision-makers need structured models when criteria conflict.

This theoretical architecture explains the research problem directly. Fragmented appraisal practice is not only a procedural weakness; it reflects an incomplete understanding of investment readiness. Financial appraisal captures repayment capacity but not Maqasid alignment. ESG disclosure captures sustainability claims but not sukuk structuring feasibility. Shariah review verifies contractual permissibility but does not always compare measurable project-level development outcomes. SC-SIIRS connects these partial lenses into one auditable decision-support architecture.

7.3 Integrated Theoretical Framework

Table 3 maps each theoretical lens to its operational role in SC-SIIRS, showing that the framework is theory-informed rather than a decorative list of concepts.

Table 3. Integrated Theoretical Architecture of SC-SIIRS

Theoretical and core source	lens	How it explains the problem	SC-SIIRS operational role	Boundary condition
Maqasid al-Shariah (Chapra, 2008; Ahmed, 2011; Dusuki & Abdullah, 2007)		Formal compliance alone cannot define Islamic investment worthiness; ethical substance, public welfare and harm reduction must be visible.	Anchors Maqasid Shariah governance, public welfare, harm reduction, and preservation of wealth, life and intellect.	Maqasid proxies can be subjective; expert validation and explicit scoring rules are required.
Stakeholder theory (Freeman, 1984)		Infrastructure projects affect users, communities, regulators and future beneficiaries, not only issuers and investors.	Anchors social inclusion, accessibility, public-benefit indicators and governance transparency.	Stakeholder theory does not determine Shariah legality; it operates under the Maqasid lens.
Sustainable finance / integrated value (Schoenmaker & Schramade, 2019)		Capital allocation should assess financial, social and environmental value together, not as disconnected reporting categories.	Anchors ESG contribution, climate resilience, environmental impact and long-term risk exposure.	ESG remains insufficient if detached from Shariah governance and Maqasid objectives.
Financial intermediation and delegated monitoring (Diamond, 1984)		Intermediaries reduce information asymmetry before capital is committed; sukuk pre-screening requires structured evidence.	Anchors financial viability, repayment coverage, documentation quality, data availability and project-risk indicators.	The theory is conventional; only its monitoring logic is adapted, not interest-based lending assumptions.
Decision-support systems theory (Simon, 1960; Keen & Scott Morton, 1978)		Bounded rationality limits unaided human comparison of conflicting criteria; a DSS structures, but does not replace, judgment.	Anchors AHP weighting, TOPSIS ranking, MEREC/COPRAS checks, auditability and human-in-the-loop governance.	The model remains advisory; Shariah boards and investment committees retain final authority.

Note. SC-SIIRS = Shariah-Compliant Smart Infrastructure Investment Readiness Score; ESG = environmental, social and governance. The table specifies how each theory constrains the model through a measurable dimension, governance rule or methodological choice.

7.4 Alignment Between Theory and Variables

The conceptual variables in SC-SIIRS are not orphan indicators. Financial viability derives from financial intermediation theory and sustainable finance because sukuk arrangers must assess repayment capacity, cash-flow resilience and capital allocation quality. Maqasid impact derives from Maqasid al-Shariah because project readiness must include public

welfare, harm reduction and ethical substance. ESG contribution derives from sustainable finance and stakeholder theory because infrastructure produces environmental and social externalities. Smart infrastructure performance derives from decision-support systems logic because technical performance indicators are treated as structured evidence for decision-making rather than as decorative smart-city metrics. Governance and transparency derive from Maqasid, stakeholder theory and decision-support systems theory because a Shariah-sensitive model must be explainable, auditable and subject to human review.

This alignment also clarifies what the model does not measure. SC-SIIRS does not measure investor sentiment, market pricing, macroeconomic sukuk spreads or post-issuance performance. These variables are excluded because they belong to pricing, market microstructure or impact evaluation studies, not to the pre-structuring readiness problem addressed here.

7.5 Alternatives Considered and Rejected

Several alternative theories were considered but not selected as primary foundations. Agency theory could explain conflicts between originators, investors and managers, but it is too narrow for this study because the core problem is not only principal-agent misalignment; it is multidimensional readiness across Shariah, sustainability, finance and infrastructure performance. Institutional theory could explain regulatory differences across jurisdictions, but the study does not primarily investigate organizational legitimacy or institutional diffusion. Transaction cost economics could explain structuring frictions, but it would understate the ethical and public-welfare dimensions central to Maqasid-based appraisal. These theories may inform the interpretation of constraints, but they do not provide the main architecture for the SC-SIIRS model.

The selected integrated framework is therefore the most suitable lens because it connects the ethical question of what should be financed, the stakeholder question of who is affected, the sustainable finance question of what value is created, the intermediation question of how uncertainty is reduced, and the decision-support question of how complex trade-offs can be made transparent.

7.6 Theoretical Propositions Guiding the Study

The framework leads to three theoretical propositions rather than formal hypotheses. First, smart infrastructure sukuk readiness is multidimensional and cannot be validly assessed

through financial viability alone. Second, Maqasid, ESG and infrastructure indicators require expert-informed weighting because they include normative and context-sensitive judgments. Third, MCDM-based decision support can improve comparability only if the scoring logic remains explainable, auditable and subordinate to human Shariah governance. These propositions guide the study questions and justify the mixed-methods design used in the study.

8. Conceptual Framework: SC-SIIRS

8.1 Definition and Function of SC-SIIRS

SC-SIIRS stands for Shariah-Compliant Smart Infrastructure Investment Readiness Score. It is not an imported off-the-shelf model. In this proposal, SC-SIIRS is the study-specific conceptual architecture used to evaluate whether a smart infrastructure project is sufficiently documented, Shariah-sensitive, financially viable, sustainable, technically credible and risk-aware to proceed toward sukuk structuring.

The model is conceptual before it is computational. Its function is to define what must be observed, measured, weighted and interpreted before an Islamic bank, sukuk issuer, Shariah board or infrastructure authority invests resources in a full structuring process. The final output is not a fatwa, credit rating or investment recommendation. It is a structured pre-screening signal that makes project trade-offs visible and auditable.

8.2 Why SC-SIIRS Is Required

Conventional financial appraisal is too narrow for this problem because it prioritizes cash-flow strength while under-specifying Maqasid impact, Shariah governance and public-welfare outcomes. ESG-only assessment is also insufficient because a project can look sustainable without having an asset structure, revenue logic or contractual profile suitable for sukuk. Smart-city readiness models add useful technical indicators, but they usually assess urban performance rather than Shariah-compliant financing readiness.

SC-SIIRS is therefore justified because it integrates the exact dimensions that the study gap identifies as fragmented: financial viability, Maqasid al-Shariah alignment, ESG contribution, smart infrastructure performance, Shariah governance and risk exposure. Its advantage is not that it replaces existing tools; its advantage is that it forces their evidence into one transparent project-level decision sequence.

8.3 Model Anatomy and Variable Logic

The framework treats the candidate smart infrastructure project as the unit of analysis. The input dimensions are the evidence categories used to assess readiness. The output is the composite SC-SIIRS readiness score and its decision band. The mediating mechanism is MCDM aggregation, which transforms weighted criteria into a comparable project ranking.

The main moderators are expert-derived weights, jurisdictional Shariah-governance expectations, data quality and project type.

The logic is evaluative rather than causal. The model does not claim that Maqasid alignment causes financial viability, or that ESG contribution causes Shariah compliance. It argues that stronger evidence across the selected dimensions increases a project's readiness classification, while weak evidence in a critical gate can reduce or block progression to sukuk structuring.

Figure 1 presents SC-SIIRS as a staged appraisal architecture. The figure should be read from left to right: project evidence enters the model, mandatory Shariah and documentation gates filter ineligible cases, eligible projects are scored across the core dimensions, and the resulting MCDM output is translated into a readiness band for human review.

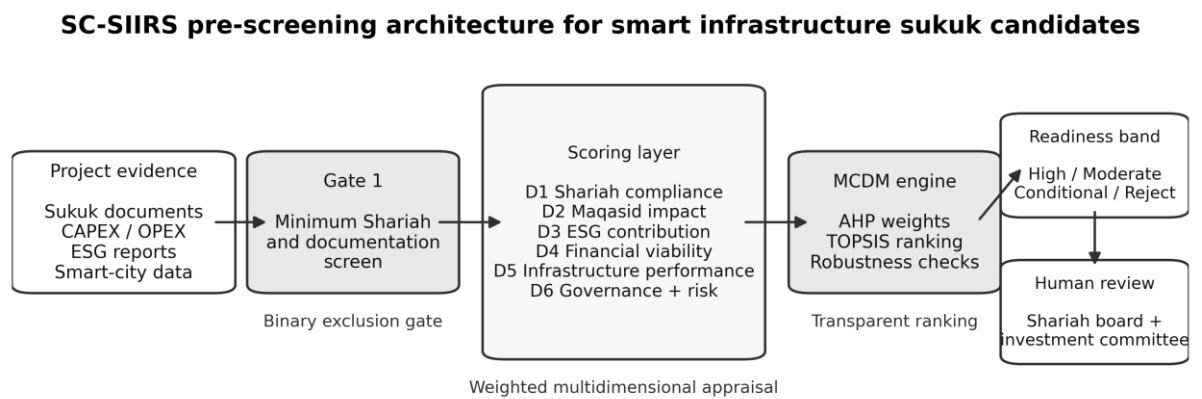


Figure 1. SC-SIIRS Conceptual Pipeline

Note. SC-SIIRS = Shariah-Compliant Smart Infrastructure Investment Readiness Score; ESG = environmental, social and governance. The pipeline shows how project evidence passes through Shariah/documentation gates before weighted scoring, MCDM ranking and human review.

Table 4 translates the SC-SIIRS concept into operational components by separating binary filters, measurable inputs, aggregation mechanisms and advisory outputs.

Table 4. Operational Architecture of the SC-SIIRS Framework

Shariah gate (binary filter)	Permissibility, asset linkage and revenue acceptability.	Sukuk documents, Shariah certificate, asset and revenue evidence.
Financial viability (input)	Capacity to sustain sukuk-related obligations.	CAPEX, OPEX, cash flow, payback, coverage ratio, NPV/IRR

				where available.
Maqasid impact (input)		Public welfare, harm reduction, equitable access and preservation objectives.		Safety, access, inclusion, environmental stewardship and expert scoring.
ESG contribution (input)		Sustainability evidence beyond simple labelling.		Emissions, resource efficiency, social inclusion, disclosure and governance quality.
Smart infrastructure performance (input)		Operational value of the smart infrastructure system.		Mobility gain, reliability, interoperability, digital access and resilience metrics.
Governance and transparency (safeguard)		Auditability of evidence, scoring and review process.		Documentation quality, explanation format, Shariah governance trail and human review.
Risk exposure (penalty input)		Factors that reduce readiness despite strong impact or return.		Regulatory, demand, technological, implementation, Shariah and reputational risk.
MCDM (mediator)	aggregation	Conversion of weighted evidence into a comparable score and ranking.		AHP weights, TOPSIS ranking, MEREC/COPRAS robustness checks.
SC-SIIRS output (dependent output)		Final readiness score and decision band for pre-structuring use.		Normalized score classified as high, moderate, conditional or not recommended.

Note. SC-SIIRS = Shariah-Compliant Smart Infrastructure Investment Readiness Score; ESG = environmental, social and governance; AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; MEREC = Method Based on the Removal Effects of Criteria; COPRAS = Complex Proportional Assessment. The framework is designed for ex-ante pre-screening and does not replace legal due diligence, credit assessment, sukuk pricing or final Shariah board approval.

8.4 Scoring Logic and Decision Bands

SC-SIIRS follows a gate-and-score logic. A project must first pass minimum Shariah and documentation gates. Only then can it be scored across the weighted dimensions. This sequence matters: a strong ESG profile or attractive financial return cannot compensate for a failed Shariah gate. Formal Shariah permissibility, in turn, does not make a project highly ready if financial viability, public-welfare value, operational performance or risk controls remain weak.

Table 5 defines how the final SC-SIIRS score is converted into an advisory readiness band for pre-structuring decisions.

Table 5. SC-SIIRS Readiness Bands and Pre-Structuring Decision Rules

85-100	High	Strong candidate for detailed structuring.	Proceed to legal, financial and Shariah due diligence.
70-84	Moderate	Viable but weak in one or two dimensions.	Proceed conditionally with targeted remediation.
55-69	Conditional	Material gaps weaken sukuk readiness.	Do not proceed until evidence is corrected.
Below 55	Not recommended	Fails minimum readiness expectations.	Reject or redesign project structure/documentation.

Note. Thresholds are provisional and will be reviewed during expert validation. The readiness band is an advisory pre-screening signal, not an investment recommendation, credit rating, legal opinion or Shariah certification.

Figure 2 visualizes the readiness bands. Its main analytical function is to prevent a false binary between accepted and rejected projects. The conditional and moderate bands are important because they identify projects that may become sukuk-ready if weaknesses in data quality, Maqasid evidence, technical performance or governance documentation are corrected.

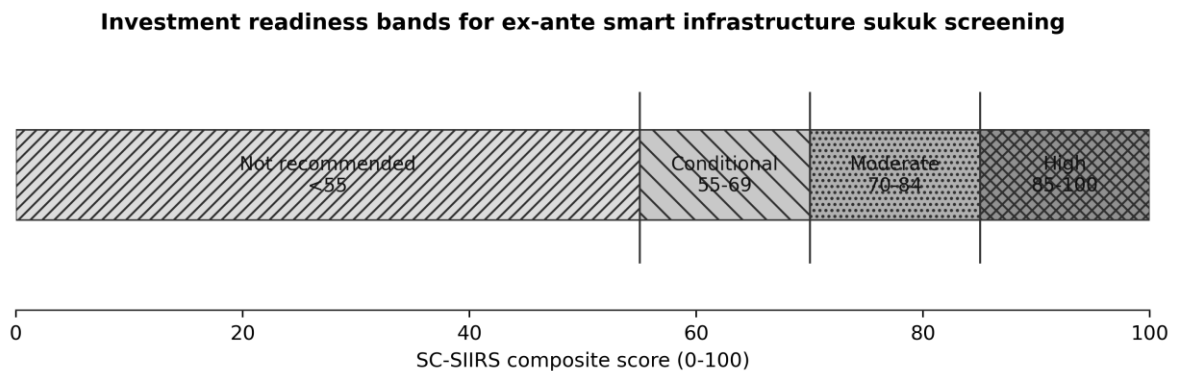


Figure 2. SC-SIIRS Readiness Bands

Note. The bands translate the final readiness score into advisory categories: not recommended, conditional, moderate and high readiness. They indicate the level of evidence correction, restructuring or Shariah-governance review required before formal sukuk structuring.

8.5 Alignment with the Research Questions

The conceptual framework mirrors the study's research questions. RQ1 is addressed by defining the dimensions and indicators required for Shariah-compliant smart infrastructure sukuk appraisal. RQ2 is addressed by translating those indicators into a weighted MCDM architecture. RQ3 is addressed by applying the score to representative project scenarios and observing how rankings change under robustness checks. RQ4 is addressed by evaluating

whether the model remains explainable, auditable and compatible with Shariah governance expectations.

This alignment prevents scope creep. The framework does not try to measure the entire Islamic finance market, certify real sukuk, forecast investor demand, or construct a machine-learning system. It is deliberately restricted to one analytical task: producing a transparent project-level readiness score for pre-structuring decision support.

9. Methodology

9.1 Methodological Positioning and Research Design

This study uses a pragmatic, sequential mixed-methods design. The quantitative strand builds SC-SIIRS; the qualitative strand tests whether the model is intelligible, defensible and usable within Shariah governance and infrastructure-finance practice. This division follows the logic of mixed-methods research: numerical modelling supports comparison and ranking, while qualitative expert interpretation captures meaning, governance judgment and contextual validation (Creswell & Plano Clark, 2018).

The research question dictates the design. It does not ask only whether one variable affects another. It asks how a transparent decision-support framework can integrate financial viability, Maqasid al-Shariah, ESG contribution, smart infrastructure performance and governance risk for sukuk pre-screening. A purely quantitative design would produce a score but under-explain Shariah acceptability and institutional usability. A purely qualitative design would generate governance insight but fail to deliver the auditable scoring architecture promised by the research aim. The mixed-methods design is therefore necessary, not decorative.

The study rejects three alternatives. A single case study is rejected because it would make the model too dependent on one jurisdiction or project. A large-scale survey is rejected because the target population is not the general public but a narrow expert group. A purely econometric design is rejected because the study does not test historical market performance; it develops and validates an ex-ante project-appraisal framework.

9.2 Unit of Analysis, Scope and Research Objects

The unit of analysis is the smart infrastructure project proposed for Shariah-compliant sukuk financing. The unit is not the Islamic bank, the investor, the city, or the sukuk market as a whole. This distinction matters because the proposed framework is designed for the pre-structuring stage, when decision-makers compare candidate projects before committing to legal structuring, Shariah approval and issuance preparation.

The empirical application will use three to five representative smart infrastructure sukuk scenarios. Eligible scenarios may include intelligent transport corridors, IoT-enabled water-loss detection systems, digital utility infrastructure, smart bus rapid transit support systems, or climate-resilient urban assets. Each scenario must contain minimum evidence on asset logic, revenue source,

CAPEX, OPEX, expected cash-flow mechanism, ESG contribution, smart infrastructure output and governance risk. Where real project data are incomplete, calibrated synthetic cases will be used and explicitly labelled.

9.3 Methodological Alignment Matrix

Table 6 shows the mirror alignment between research questions, evidence, methods and outputs; every analytical tool is tied to a defined methodological function.

Table 6. Methodological Alignment Matrix

Research question	Evidence required	Method	Expected output
RQ1. Which criteria are required for SC-SIIRS?	Literature, standards, sukuk infrastructure and ESG indicators	Document analysis and criteria mapping	Validated SC-SIIRS criteria hierarchy
RQ2. How should criteria be weighted?	Expert pairwise-comparison judgments	AHP weighting with consistency screening	Expert-derived dimension and sub-criterion weights
RQ3. How do candidate projects rank?	Scenario-level financial, ESG, Maqasid, technical and risk data	TOPSIS ranking; normalized scoring matrix	Project rankings and readiness bands
RQ4. How robust and usable is the model?	Alternative weights/rankings and expert interpretation	MEREC, COPRAS, sensitivity analysis and thematic analysis	Robustness diagnosis and governance-validity assessment

Note. AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; MEREC = Method Based on the Removal Effects of Criteria; COPRAS = Complex Proportional Assessment. The table separates model construction, weighting, ranking and validation so that each method has a defined role.

9.4 Population, Sampling and Recruitment

The expert population comprises professionals and academics with demonstrable expertise in Islamic finance, sukuk structuring, Shariah governance, sustainable finance, ESG assessment, infrastructure finance, public-private partnerships or smart-city investment appraisal. The study will use purposive sampling because the study requires specialized judgement rather than statistical representativeness.

The minimum viable sample is 8-12 valid AHP expert responses. An aspirational target of 15-20 responses will be pursued if access allows. A smaller but qualified panel is preferable to a larger but weakly informed sample because AHP depends on informed pairwise judgement. The qualitative phase will use 6-10 semi-structured interviews selected from the same expert domains, with priority given to participants able to comment on Shariah governance, practical adoption and interpretability.

Table 7 specifies the minimum evidence streams and expert inputs required to make the methodology executable within master-level time and access constraints.

Table 7. Sampling and Data-Collection Plan

Data stream		Target source		Minimum		Recruitment /		Function in	
		or participant	target			collection logic	the study		
Secondary project evidence		Public	sukuk	3-5	project	Desk-based		Populates	SC-
		prospectuses, development-bank reports, ESG disclosures, transport authority documents	scenarios			document collection and scenario calibration	SIIRS project matrix		
AHP expert survey		Islamic		8-12	valid	Purposive		Derives	
		finance, governance, infrastructure experts	Shariah and responses			recruitment through professional and academic networks	criterion weights		
Semi-structured interviews		Selected		6-10		Purposive		Validates	
		experts with governance or implementation insight	interviews			follow-up after AHP and document analysis	usability, explainability and Shariah-governance fit		

Note. The sample is specialized rather than statistically representative. The validity claim is expert-informed decision support, not population generalization.

9.5 Data Collection Instruments

Three instruments will be used. First, a secondary-data extraction matrix will record each scenario's financial, ESG, Maqasid, technical, governance and risk indicators. Second, an AHP questionnaire will ask experts to compare SC-SIIRS dimensions and selected sub-criteria using pairwise comparisons. Third, a semi-structured interview guide will test whether the resulting framework is understandable, institutionally usable and compatible with Shariah governance.

The AHP questionnaire and interview guide will be developed in three steps: literature-derived drafting, expert review by at least one academic or practitioner, and a pilot test with two to three participants. The pilot will check wording clarity, completion time, respondent burden and the risk of inconsistent interpretation of terms such as Maqasid impact, Shariah governance, readiness score and risk exposure.

The full SC-SIIRS criteria and indicator matrix is provided in Appendix A; the secondary data extraction template is provided in Appendix B; the AHP expert pairwise comparison questionnaire is provided in Appendix C; and the semi-structured interview guide is provided in Appendix D.

9.6 Quantitative Data Analysis Plan

The quantitative analysis will proceed in four stages. First, criteria will be organized into a hierarchy: goal, dimensions, sub-criteria, indicators and scoring rules. Second, AHP will derive expert-based weights. Individual pairwise-comparison matrices will be screened for consistency using the consistency ratio; highly inconsistent responses will be reviewed, revised where possible, or excluded from the core weighting set. Third, TOPSIS will rank the candidate project scenarios by distance from ideal and anti-ideal project profiles. This is appropriate because SC-SIIRS contains beneficial criteria, such as ESG contribution and Maqasid impact, and non-beneficial criteria, such as risk exposure and implementation complexity.

Fourth, robustness will be tested through MEREC, COPRAS and sensitivity analysis. MEREC will test whether objective criterion-importance patterns diverge from expert-derived AHP weights. COPRAS will test whether ranking conclusions remain stable under an alternative MCDM aggregation logic. Sensitivity analysis will vary key weights, especially Maqasid, financial viability and risk exposure, to determine whether project rankings are stable or fragile.

9.7 Financial and Non-Financial Scoring Rules

Financial indicators will include CAPEX, OPEX, expected net operating cash flow, payback period, NPV or equivalent project-value proxy, IRR where applicable, and sukuk distribution coverage logic. Non-financial indicators will include Maqasid contribution, ESG performance, infrastructure performance, governance transparency, data quality and risk exposure. All indicators will be normalized before aggregation. The scoring rules will distinguish between benefit criteria, where higher values improve readiness, and cost/risk criteria, where higher values reduce readiness.

The model will not treat all criteria as equally measurable. Financial indicators can often be measured numerically; Maqasid and governance indicators require structured proxies and expert judgement. This is why AHP is used for weighting and interviews are used for governance validation.

9.8 Qualitative Data Analysis Plan

Interview data will be analysed using thematic analysis. Thematic analysis is appropriate because it offers a flexible and systematic way to identify patterns across expert accounts without forcing the data into a purely statistical structure (Braun & Clarke, 2006). The coding process will follow six steps: familiarization, initial coding, theme development, theme review, theme definition and final interpretation.

Coding will be partly deductive and partly inductive. Deductive codes will be based on the SC-SIIRS dimensions: Shariah governance, Maqasid validity, ESG credibility, financial appraisal, infrastructure performance, risk exposure and institutional usability. Inductive coding will capture

unexpected concerns, such as regulatory constraints, data-quality doubts, concerns about algorithmic opacity or disagreement over Maqasid proxies. NVivo, Atlas.ti or a documented Excel coding matrix will be used depending on software access.

9.9 Integration of Quantitative and Qualitative Findings

Integration will occur at the interpretation stage. Quantitative analysis will show how projects rank and whether rankings remain stable under robustness tests. Qualitative analysis will explain whether experts consider those rankings understandable, ethically defensible and institutionally usable. The logic is sequential: the model produces a ranking; experts interrogate the ranking logic; the final interpretation reports both the numerical result and the governance conditions under which the result can be trusted.

9.10 Reproducibility, Validity and Bias Controls

Table 8 summarizes the controls used to reduce subjectivity, document assumptions and prevent the scoring model from becoming a black box.

Table 8. Reproducibility and Validity Controls

Threat	Control procedure	Residual limitation
Unclear criteria	Criteria dictionary, literature mapping and pilot review	Some constructs remain proxy-based
Expert inconsistency	AHP consistency screening and respondent revision where feasible	Expert judgement remains subjective
Ranking instability	MEREC, COPRAS and sensitivity analysis	Different MCDM logics may produce different rankings
Confirmation bias in interviews	Semi-structured guide, audit trail and inclusion of negative cases	Researcher interpretation cannot be fully eliminated
Weak scenario data	Minimum data threshold, explicit synthetic-case labelling and conservative assumptions	Real transaction-level data may remain unavailable
Overclaiming Shariah validity	SC-SIIRS framed as decision support, not certification or fatwa	Final Shariah judgement remains outside the model

Note. The methodology does not eliminate uncertainty. It makes assumptions explicit, tests model sensitivity and separates computational ranking from final human judgment.

9.11 Methodology Flowchart

Figure 3 summarizes the operational sequence of the study. It also clarifies that the methodology is not a loose collection of tools: each stage produces the input required by the next stage.

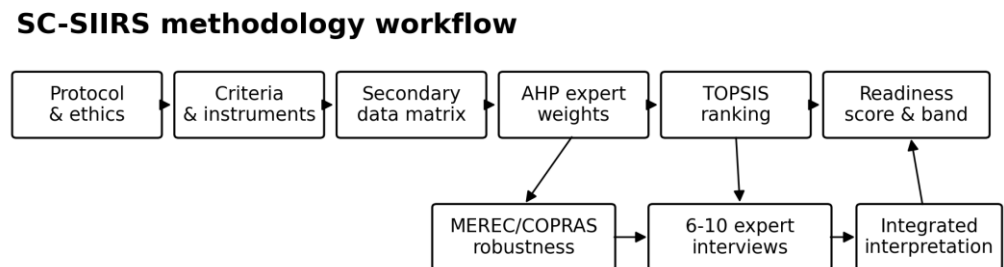


Figure 3. SC-SIIRS Research Methodology Workflow

Note. AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; MEREC = Method Based on the Removal Effects of Criteria; COPRAS = Complex Proportional Assessment. The workflow links protocol approval, instrument development, weighting, ranking, robustness checks, qualitative validation and final reporting.

9.12 Methodological Boundary Statement

This methodology is feasible because it refuses to do everything. It does not estimate market-wide sukuk performance, issue a legally binding Shariah ruling, price a sukuk transaction or claim causal impact. It develops and validates a project-level decision-support framework. Its strongest validity claim is not universal prediction; it is transparent, reproducible and Shariah-sensitive pre-screening of candidate smart infrastructure sukuk projects.

10. Preliminary Results and Results Reporting Plan

10.1 Status of Results at Proposal Stage

At the proposal stage, this study claims no empirical findings. This section therefore defines the future reporting architecture rather than presenting completed evidence. The distinction matters. The proposal can specify what will be measured, ranked, compared and validated; it cannot report project rankings or interview themes before data collection is complete.

The eventual results section will follow the methodology's sequence: sample and data completeness, AHP weighting outputs, TOPSIS project rankings, MEREC and COPRAS robustness checks, sensitivity analysis, qualitative validation themes and mixed-methods integration. This order reduces selective reporting and keeps the results tied to the research questions instead of turning them into a catalogue of technical outputs. The logic also aligns with transparent reporting expectations in APA Journal Article Reporting Standards and EQUATOR Network guidance on complete and structured research reporting (American Psychological Association, n.d.; EQUATOR Network, n.d.).

10.2 Quantitative Results to be Reported

The quantitative results will begin with a concise description of the expert panel and scenario dataset. The text will report the number of valid AHP responses, expert categories represented, rejected or revised matrices, missing project indicators and the proportion of synthetic versus real data used in each scenario. These details will be reported before any ranking results because the credibility of SC-SIIRS depends on the quality of the input evidence.

The first analytical output will be the final AHP weight vector for the SC-SIIRS dimensions. The text will identify only the highest-weighted and lowest-weighted dimensions and will direct the reader to the weighting table for full values. The results will not claim that one dimension is intrinsically more important than another in Islamic finance; it will only report how the recruited expert panel weighted the dimensions within this specific study.

The second analytical output will be the TOPSIS ranking of the three to five smart infrastructure sukuk scenarios. The text will identify the top-ranked, lowest-ranked and borderline projects, while avoiding repetition of every score already displayed in the table or

figure. The ranking will be presented as an ex-ante screening result, not as a final investment recommendation or Shariah approval.

The third analytical output will be the robustness analysis. MEREC will test whether objective criterion-weight sensitivity materially differs from AHP expert weights. COPRAS will test whether project rankings remain stable under an alternative aggregation method. Figure 4 will show rank stability, reversals and sensitivity to plus/minus 20 percent changes in key weights.

10.3 Qualitative Results to be Reported

The qualitative results will be organized by themes rather than by interviewee. The expected theme structure will include framework validity, Maqasid proxy acceptability, Shariah governance compatibility, data-access constraints, institutional adoption barriers and safeguards against greenwashing or Shariah-washing. Representative quotations will be used sparingly and only when they clarify a theme that cannot be adequately captured through numerical scoring.

The qualitative section will not be used to praise the framework or defend the researcher's assumptions. It will report convergence, disagreement and unresolved objections. If experts reject a Maqasid proxy, question the use of numerical scoring, or raise concerns about algorithmic overreach, those findings will be reported explicitly rather than buried in limitations.

10.4 Integration of Quantitative and Qualitative Results

The integrated results will compare the MCDM ranking pattern with expert judgments from the interviews. Agreement between the quantitative ranking and qualitative validation will be treated as evidence of practical plausibility. Disagreement will be treated as analytically valuable because it can reveal criteria that are mathematically influential but institutionally or Shariah-governance sensitive.

This integration will be reported in a restrained manner. The results section will state what the model produced and what experts accepted, questioned or rejected. Interpretation of why those patterns matter for Islamic finance theory, sukuk governance or sustainable infrastructure policy will be reserved for the discussion section.

10.5 Treatment of Negative, Unstable or Unexpected Results

The results chapter will report negative and unstable results as primary evidence, not as methodological embarrassment. Possible negative results include low expert agreement, high AHP inconsistency, unstable TOPSIS/COPRAS rankings, weak Maqasid scores for financially attractive projects, or strong ESG performance combined with poor Shariah-structuring feasibility. These outcomes would not invalidate the study; they would identify where a pre-screening framework is most needed.

If no project achieves a high readiness band, the result will be reported directly. If the highest-ranked project changes under sensitivity analysis, the result will be reported as ranking instability. If qualitative experts reject the operationalization of one dimension, the result will be reported as a framework revision requirement rather than ignored.

10.6 Results Reporting and Visual Integration

The results narrative will not reproduce numerical tables line by line. Complete values will be reported in the relevant tables, while the text will identify the dominant pattern, the most material trade-off, and any unstable or contradictory result. This keeps the results section analytical rather than repetitive.

Figure 4 provides the reporting template for the robustness analysis. It is included to show how ranking stability, model disagreement and sensitivity to weight changes will be reported without overclaiming statistical certainty.

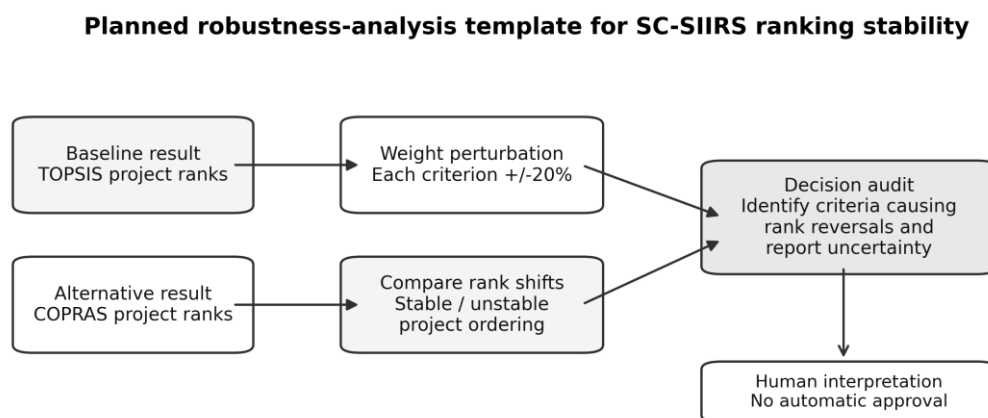


Figure 4. SC-SIIRS Robustness Reporting Template

Note. SC-SIIRS = Shariah-Compliant Smart Infrastructure Investment Readiness Score; AHP = Analytic Hierarchy Process; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; COPRAS = Complex Proportional Assessment. The template separates baseline rankings, alternative rankings and sensitivity bands so unstable scenarios are not presented as definitive results.

11. Risk Management and Contingency Plan

This section treats feasibility as a managed design condition, not as an optimistic assumption. The risk plan follows a risk-register logic: identify the threat, assess probability and impact, define preventive mitigation, specify a contingency trigger and state the residual effect on the study. This format is consistent with research-funding expectations that risk information should be structured enough to support communication and management decisions (Science and Technology Facilities Council, 2021).

The project is feasible because no single data source or expert group is treated as a fatal dependency. The core contribution is the design and validation of a transparent project-level decision-support framework; therefore, the study can remain valid through calibrated public cases, reduced expert samples, sensitivity analysis and clearly disclosed assumptions if ideal access conditions are not achieved.

11.1 Risk Assessment Logic

Each risk is assessed through two dimensions: probability and impact. Probability indicates how likely the risk is to occur during the study period; impact indicates how severely it would affect completion, validity or credibility. The highest-priority risks are threats that could damage the study design: insufficient expert recruitment, incomplete sukuk or infrastructure data, inconsistent AHP responses, contested Maqasid proxies, ranking instability and regulatory changes affecting sukuk governance assumptions.

Risk responses are separated into mitigation and contingency. Mitigation refers to preventive actions taken before the risk materializes. Contingency refers to the alternative procedure activated if the risk occurs. This distinction prevents the plan from becoming a passive list of worries.

11.2 Risk Register and Contingency Dashboard

Table 9 translates the main feasibility threats into concrete triggers and contingency actions, separating strategic research risks from routine operational inconvenience.

Table 9. Research Risk Register and Contingency Responses

Risk and rating	Trigger / warning sign	Mitigation and contingency response
Limited access to real sukuk or smart infrastructure project data. Probability: high. Impact: high.	Prospectuses, project documents or financial details are incomplete, confidential or non-comparable.	Mitigation: prioritize public sukuk prospectuses, IsDB/World Bank documentation, transport authority reports and sustainability disclosures. Contingency: use three to five calibrated public or synthetic scenarios. Residual effect: the study remains a framework-validation exercise, not a proprietary transaction valuation.
Insufficient expert recruitment for AHP weighting. Probability: medium-high. Impact: high.	Fewer than 8 valid expert responses after two recruitment rounds.	Mitigation: recruit across four pools: Islamic finance, Shariah governance, ESG/sustainable finance and infrastructure/PPP specialists. Contingency: use a reduced expert panel with transparent limitations; strengthen sensitivity analysis and report weight dispersion.
Inconsistent AHP pairwise judgments. Probability: medium. Impact: medium-high.	Consistency ratio exceeds the accepted threshold for several matrices.	Mitigation: provide precise criterion definitions, pilot the questionnaire and allow one revision. Contingency: exclude unrevised inconsistent matrices from the core model and retain them only for sensitivity discussion.
Contested Maqasid indicators or weak construct validity. Probability: medium. Impact: high.	Experts disagree on whether a proxy genuinely represents preservation of life, wealth, intellect, welfare or harm reduction.	Mitigation: define each proxy before data collection and map it to Maqasid literature, ESG logic and project-level evidence. Contingency: run alternative weighting scenarios and mark contested indicators as provisional.
Divergent TOPSIS and COPRAS rankings. Probability: medium. Impact: medium.	The top-ranked project changes across ranking methods or under +/-20 percent weight variation.	Mitigation: predefine TOPSIS as the primary ranking method and COPRAS as a robustness check. Contingency: report divergence as model sensitivity; classify borderline projects as conditional cases requiring qualitative review.

Risk and rating	Trigger / warning sign	Mitigation and contingency response
Regulatory or Shariah-standard interpretation changes during the study. Probability: medium. Impact: high.	AAOIFI, IFSB or market guidance changes the interpretation of sukuk ownership, asset transfer or governance requirements.	Mitigation: treat standards as time-stamped governance inputs and avoid claiming universal legal finality. Contingency: add a version-control note; results apply under cited standards and require recalibration if the governance baseline changes.
Ethics approval or consent procedures delay expert interviews. Probability: medium. Impact: medium.	Approval or participant consent is not secured before the planned interview window.	Mitigation: prepare information sheet, consent form, anonymization protocol and data-storage plan early; ethics self-assessment guidance emphasizes identifying ethics issues before project execution (European Commission, 2021). Contingency: proceed with desk-based scenario modelling and conduct interviews only after approval.
Overclaiming AI capability or Shariah authority. Probability: medium. Impact: high.	Reviewers interpret SC-SIIRS as an autonomous AI system or a substitute for Shariah board judgment.	Mitigation: use precise language: MCDM-based decision-support framework, not automated fatwa, certification or legal approval. Contingency: revise outputs to show advisory readiness bands, evidence traceability and human-in-the-loop Shariah governance.
Negative, unstable or non-confirmatory findings. Probability: medium. Impact: medium.	No project reaches a high-readiness band; expert weights are dispersed; financial and Maqasid scores conflict.	Mitigation: state in advance that instability and disagreement are valid findings. Contingency: publish the negative result as evidence of why integrated screening is necessary; the contribution becomes diagnostic rather than confirmatory.

Note. Probability and impact are assessed qualitatively because this is a proposal-stage risk register, not an executed project-monitoring report.

11.3 Worst-Case Scenario and Scientific Value of Negative Results

The most damaging scenario would combine three failures: limited access to real sukuk project data, a small expert sample and unstable MCDM rankings. Even under that scenario, the project would not collapse if its scope is held correctly. The thesis would shift from claiming empirical generalizability to demonstrating a transparent framework, testing its behaviour under constrained information and documenting where smart infrastructure sukuk appraisal remains institutionally under-specified.

Negative or unstable findings will be treated as evidence rather than failure. If no project reaches the high-readiness band, the result will show that apparent eligibility is not

equivalent to investment readiness. If financially strong projects score weakly on Maqasid or governance indicators, the model will reveal a substantive trade-off that conventional appraisal would likely miss. If expert disagreement is high, the finding will identify which dimensions require better standardization before Islamic financial institutions can adopt comparable screening tools.

11.4 Decision Rules for Activating Contingency Plans

- If fewer than 8 valid AHP responses are collected, the study will report a reduced expert-weight model and expand sensitivity analysis.
- If fewer than three real project cases are usable, the study will use calibrated synthetic cases and label them explicitly.
- If a Maqasid proxy is rejected by multiple experts, it will be downgraded, revised or treated as contested rather than silently retained.
- If ranking results are unstable across TOPSIS, COPRAS or weight variation, the study will classify the affected projects as conditional rather than presenting a false definitive order.
- If interview access is delayed by ethics or recruitment constraints, the quantitative model will proceed, while the qualitative phase will be limited to validation rather than broad institutional inference.

11.5 Residual Risk Statement

After mitigation, the main residual risks are limited generalizability and proxy contestability. These risks are acceptable because the project is positioned as a design and validation study, not as a universal rating system for all sukuk markets. The final output will therefore present SC-SIIRS as a transparent pre-screening architecture that requires jurisdictional calibration, expert validation and Shariah-board oversight before institutional adoption.

12. Ethical Considerations and Data Protection

This section treats ethics and data protection as design controls rather than administrative formalities. The study involves adult expert participants, professional opinions, AHP pairwise-comparison responses and semi-structured interview data; therefore, the central risks are not physical harm, but reputational exposure, deductive identification, coercive recruitment, loss of confidentiality and misuse of identifiable data. The protocol is designed around autonomy, non-maleficence, beneficence and justice, the four-principles framework associated with Beauchamp and Childress (2019), and around the study-ethics duties of respect for persons, beneficence and justice articulated in the Belmont Report (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979).

12.1 Ethical Governance and Prior Approval

No primary data will be collected before the project has been reviewed and approved, or formally exempted, by the relevant institutional ethics committee or research review board. Recruitment emails, information sheets, consent forms, interview guides and the AHP questionnaire will be submitted for review before circulation. If the institution requires data-protection review by a data protection officer or equivalent authority, that review will be completed before any personal data are processed.

The participant information sheet is provided in Appendix E, the informed consent form is provided in Appendix F, the data management and anonymization protocol is provided in Appendix G, and the ethics approval status statement is provided in Appendix H.

The study will be presented to participants as a low-risk academic project, not as consultancy, investment advice, Shariah certification, legal structuring or institutional audit. This distinction matters because participants may hold senior professional roles; their views must not be misrepresented as official positions of their employers, Shariah boards or regulatory bodies.

12.2 Participants, Vulnerability Assessment and Recruitment

The participant population will consist exclusively of adults aged 18 or above with professional or academic expertise in Islamic finance, sukuk structuring, Shariah governance, ESG/sustainable finance, smart infrastructure, public-private partnerships or MCDM-based decision support. Minors will not be recruited. Because the study does not require vulnerable

populations, individuals unable to provide informed consent, dependants of the researcher, patients, students under direct assessment by the researcher or economically coerced participants will be excluded.

Recruitment will be purposive and professional. Potential participants will be contacted through institutional email, professional networks or academic channels. Recruitment messages will avoid pressure, urgency, promises of advantage or language implying that participation is expected. Participation will be voluntary, and refusal will have no academic, professional or institutional consequence.

12.3 Informed Consent and Right of Withdrawal

Before participation, each expert will receive an information sheet explaining the purpose of the study, the type of data collected, expected time commitment, recording procedures, potential risks, expected benefits, confidentiality measures, data-retention period, contact details and the right to withdraw. Consent will be documented in writing or through a recorded verbal consent procedure approved by the ethics committee. HHS/OHRP guidance emphasizes that human subjects may not be involved in covered research unless legally effective informed consent has been obtained and documented in line with applicable requirements (U.S. Department of Health and Human Services, n.d.).

Participants may withdraw from the study at any time without penalty. They may request withdrawal of their identifiable interview data until the point at which the transcript has been anonymized and integrated into aggregated analysis. After full anonymization and aggregation, withdrawal of a specific contribution may no longer be technically possible because the data will no longer be linkable to the participant. This limit will be stated clearly in the consent documentation.

12.4 Confidentiality, Anonymization and Deductive Disclosure Control

The study will use pseudonyms such as Expert 01, Expert 02 and Expert 03. Names, email addresses, institutional affiliations and signed consent forms will be stored separately from research responses. Direct identifiers will be removed during transcription. Interview quotations will be used only when they do not reveal the participant, their institution or a unique professional position.

Full anonymity cannot be guaranteed absolutely because the expert community in Islamic finance, Shariah governance and sukuk structuring is relatively small. The protocol

therefore focuses on confidentiality and deductive disclosure control. Job titles will be generalized, institutions will not be named unless explicit written permission is given, and combinations of role, jurisdiction and specialist expertise will be blurred where they could identify a participant.

12.5 Data Protection and Secure Data Management

The study will follow data minimization. Only data necessary for recruitment, consent, expert weighting, interview analysis and auditability will be collected. Personal data will not be used to make decisions about participants. The European Commission states that the GDPR applies to personal data regardless of the technology used and regardless of whether storage is digital, paper-based or otherwise organized (European Commission, n.d.). Accordingly, identifiable data will be processed only for research purposes, stored securely and reported only in anonymized or aggregated form.

Data will be stored in an encrypted, password-protected university-approved storage environment wherever available. If temporary local storage is unavoidable, files will be encrypted, protected by strong passwords and removed from local devices after secure upload. Consent forms, contact details and raw interview files will be kept separately from anonymized transcripts and scoring datasets. Public cloud platforms, personal messaging apps and unapproved AI transcription tools will not be used for identifiable research data unless explicitly authorized by the institution and compliant with applicable data-protection rules.

Table 10 identifies the ethics and data-protection risks specific to this study and links each risk to a concrete safeguard.

Table 10. Ethics and Data-Protection Risk Dashboard

Risk area	Potential harm	Control measure
Coercive recruitment	Experts may feel pressured to participate because of institutional, academic or professional relationships.	Use neutral invitation language; state that participation is voluntary; avoid recruitment through direct authority relationships; record refusal as non-response without follow-up pressure.
Deductive identification	A participant may be identifiable through a combination of role, jurisdiction, institution and specialist expertise.	Use pseudonyms; generalize role descriptions; avoid naming institutions; review quotations for identifiability before publication.

Risk area	Potential harm	Control measure
Reputational or professional exposure	Critical comments about sukuk governance, Shariah boards, ESG claims or institutional practice may be sensitive.	Report findings in aggregate; avoid attributing controversial claims to identifiable individuals; allow participants to decline recording or quotation.
Loss of confidentiality	Unauthorized access to raw interview recordings, transcripts, AHP matrices or consent forms could expose personal data.	Encrypt files; separate identifiers from responses; restrict access to the researcher and supervisor; use university-approved storage; delete local temporary copies.
Misuse of AI or third-party tools	Uploading identifiable transcripts to uncontrolled AI, transcription or cloud tools may breach confidentiality.	Do not upload identifiable data to external AI tools unless institutionally approved; anonymize before any software-assisted analysis; document any tool used.
Cross-border data transfer	Storage or processing outside the approved jurisdiction may create legal uncertainty.	Prefer EU/EEA or institution-approved storage; avoid unapproved international transfers; consult institutional data-protection guidance when necessary.
Algorithmic overclaiming	SC-SIIRS could be misread as automated Shariah approval or investment advice.	State that SC-SIIRS is advisory decision support only; final judgment remains with Shariah boards, investment committees and legal reviewers.
Data retention drift	Data may be retained longer than necessary or reused outside the consented scope.	Define a retention period consistent with institutional policy; destroy identifiers after the retention period; use future data reuse only if ethically approved and consent-compatible.

Note. The dashboard focuses on risks specific to this study. Generic operational problems are excluded unless they affect participant protection, confidentiality, data integrity or responsible interpretation of SC-SIIRS.

12.6 Data Retention, Destruction and Breach Response

Raw recordings will be retained only as long as necessary for transcription and verification. Anonymized transcripts, coded themes, aggregated AHP weights and scoring matrices will be retained according to institutional policy, normally for a defined research-retention period. After that period, identifiable data will be securely deleted. Consent forms may be retained separately for audit purposes if required by the institution.

If a suspected data breach occurs, the researcher will immediately stop further processing, secure the affected device or account, change relevant credentials, notify the supervisor and institutional data-protection or ethics contact, assess whether identifiable personal data were exposed, and follow institutional procedures for participant notification where risk is material. A breach log will document the incident, affected files, actions taken

and residual risk. The objective is not to promise that breaches are impossible, but to ensure that access is minimized, damage is contained and reporting is prompt.

12.7 Ethical Use of SC-SIIRS Outputs

SC-SIIRS will not generate binding investment recommendations, Shariah certification, credit ratings or legal opinions. It will produce an auditable readiness score and decision band for research purposes. This limitation protects participants and users from overreliance on a model that is still exploratory. The outputs will be framed as structured evidence for human review, not as automated approval.

The model will also avoid individual-level profiling. Its data concern projects, criteria, weights and expert judgments; it does not rank participants or evaluate identifiable persons. Where expert disagreement appears in the results, it will be reported as a property of the decision problem, not as evidence that a specific participant or institution is wrong.

12.8 Residual Ethical Risk Statement

After mitigation, the main residual ethical risks are deductive identification and interpretive misuse of the model. These risks are manageable because the participant population is professional and adult, the study does not involve vulnerable groups, data collection is limited to expert judgment, and all outputs are anonymized, aggregated and explicitly framed as decision support. The protocol is therefore ethically proportionate to the risk level of the study while remaining strict enough for institutional review.

13. Expected Contributions and Beneficiaries

This section defines the expected value of the study in terms of research output, beneficiary use, and transmission pathway. The claim is intentionally precise: the study does not promise market transformation by itself. It produces a reusable decision-support architecture that makes smart infrastructure sukuk appraisal more transparent, comparable and auditable at the pre-structuring stage.

13.1 Core Contribution: From Conceptual Alignment to Operational Screening

The central contribution of this research is to move Islamic sustainable finance from broad compatibility claims toward operational project appraisal. Existing debates frequently state that Islamic finance, Maqasid al-Shariah and sustainability are aligned. This study translates that alignment into a structured readiness score that can compare candidate smart infrastructure projects before sukuk structuring begins.

In five years, the value of this study should be visible in a more disciplined research vocabulary and a more practical appraisal template. Instead of asking only whether a project is Shariah-compliant or sustainability-labelled, researchers and practitioners will be able to ask whether it is financially viable, Maqasid-aligned, ESG-relevant, technically credible, governance-ready and risk-adjusted enough to proceed to sukuk structuring.

13.2 Theoretical Contributions

The study makes three theoretical contributions. First, it extends Islamic sustainable finance literature by shifting the unit of analysis from the institution or market to the individual infrastructure project. This is important because capital allocation decisions are made project by project, before sukuk documentation, Shariah approval and investor marketing.

Second, the study operationalizes Maqasid al-Shariah as an appraisal dimension rather than treating it as an abstract ethical vocabulary. The framework does not reduce Maqasid to mechanical numbers; it makes the selected proxies explicit, contestable and reviewable. This strengthens theoretical transparency because the ethical assumptions behind the score become visible instead of remaining rhetorical.

Third, the study connects decision-support systems theory with Shariah-governed finance. It shows how multi-criteria decision-making can support human judgment without

replacing Shariah boards, investment committees or legal structuring teams. The theoretical value lies in defining the boundary between algorithmic assistance and accountable human governance.

13.3 Methodological and Technical Contributions

The methodological contribution is the design of SC-SIIRS as a transparent project-level scoring architecture. AHP provides expert-derived weights, TOPSIS produces the primary ranking of project alternatives, and MEREC/COPRAS operate as robustness checks. This hierarchy prevents methodological overclaiming: the model does not rely on a single algorithm, but it also avoids pretending that method multiplication is itself a contribution.

Technically, the study delivers a replicable criteria matrix, a readiness-band logic, a sensitivity-analysis procedure and an expert-validation protocol. These outputs can be reused, revised or stress-tested by future researchers working on Islamic infrastructure finance, sustainable sukuk, Shariah governance or smart-city investment appraisal.

13.4 Practical and Policy Contributions

The practical contribution is a pre-screening tool for institutions that must decide which projects deserve further legal, financial and Shariah structuring effort. Islamic banks can use the framework to compare project candidates before allocating due-diligence resources. Sukuk issuers can use it to prepare clearer evidence packages for investors and Shariah reviewers. Shariah boards can use it to inspect trade-offs between formal compliance, public welfare, sustainability claims and risk exposure.

The policy contribution is equally concrete. Regulators and standard-setting bodies can use the framework as a basis for more structured disclosure expectations around smart infrastructure sukuk. Multilateral development institutions can adapt the criteria matrix to screen projects that combine climate relevance, real-asset financing and public welfare objectives. Infrastructure planners can use the model earlier in project design to make technically attractive projects more finance-ready.

13.5 Key Beneficiaries and Transmission Pathways

Table 11 narrows the contribution claim by identifying the institutional users of SC-SIIRS and the pathway through which each group can use the framework.

Table 11. Institutional Beneficiaries and Use Pathways for SC-SIIRS

Beneficiary group	Expected use and transmission pathway
Islamic banks and Islamic windows	Use SC-SIIRS as an early-stage screening template to compare smart infrastructure project candidates before committing due-diligence and sukuk structuring resources. Transmission: criteria matrix, readiness-band template and practitioner scoring sheet.
Sukuk issuers and special-purpose vehicles	Use the framework to prepare evidence packages that connect asset structure, revenue logic, ESG claims and Maqasid contribution. Transmission: project evidence checklist, proposal annex and investor-facing summary.
Shariah boards and governance units	Use the score as structured evidence for human review of trade-offs between formal compliance, public welfare, sustainability claims and governance risk. Transmission: audit trail, safeguards, interview findings and limitations statement.
Regulators and standard setters	Use the model as a reference for voluntary disclosure expectations and appraisal guidelines in sustainable infrastructure sukuk. Transmission: policy recommendations, standards-oriented brief and academic publication.
Multilateral development banks and public finance institutions	Adapt the framework as a project-pipeline screening tool for infrastructure, climate relevance and Shariah-compatible financing potential. Transmission: impact brief, scenario results and replicable scoring rubric.
Smart-city planners and PPP units	Use the framework during project preparation to detect weak financial, ESG or governance dimensions before financing discussions begin. Transmission: project design checklist, sensitivity-analysis outputs and readiness-band interpretation guide.
Researchers in Islamic finance, sustainable finance and MCDM	Reuse, critique or extend the framework in comparative, country-specific or sector-specific studies. Transmission: peer-reviewed article, open criteria matrix and methodological appendix.

Note. SC-SIIRS is intended for institutional pre-screening and research use. It is not designed for retail investors, does not certify Shariah compliance, and does not replace credit analysis, legal documentation, investment committee judgment or Shariah board approval.

13.6 Impact Pathway and Deliverables

The study will transmit its contribution through four concrete outputs: (1) the SC-SIIRS conceptual framework; (2) a criteria and scoring matrix; (3) a ranked application to three to five representative smart infrastructure sukuk scenarios; and (4) governance recommendations explaining how the score should be interpreted by human decision-makers.

These outputs give the study a pathway beyond the thesis document: they can become a journal article, an executive policy note, a practitioner checklist and a methodological appendix for future empirical work.

The strongest short-term outcome is improved comparability. The strongest medium-term outcome is a reusable appraisal protocol for future Islamic sustainable infrastructure finance studies. The long-term value is a shift from rhetorical compatibility between Islamic finance and sustainability toward evidence-based capital allocation for Shariah-compliant infrastructure investment.

13.7 Boundary of the Contribution Claim

The contribution is substantial but bounded. This study does not claim to transform the sukuk market by itself, predict investor demand, create a binding regulatory standard, automate Shariah certification or prove that one infrastructure technology is universally superior. Its contribution is to build and validate a disciplined appraisal architecture. That architecture is valuable precisely because it makes assumptions, trade-offs and weaknesses visible before institutions commit capital, reputation and legal resources to sukuk structuring.

14. Limitations and Delimitations

This section separates methodological delimitations from study limitations. Delimitations are the boundaries deliberately chosen by the researcher to keep the study focused; limitations are constraints that may affect interpretation despite the controls built into the study design (Theofanidis & Fountouki, 2018; Ross & Bibler Zaidi, 2019). This distinction matters because the proposal is not claiming universal Shariah certification or market-wide prediction. It is developing a bounded, auditable pre-screening framework for smart infrastructure sukuk.

14.1 Delimitations of the Study

Unit of analysis. The study is delimited to smart infrastructure projects that could plausibly be financed through Shariah-compliant sukuk. It does not evaluate Islamic banks as whole institutions, national sukuk markets or city-level smart-city maturity.

Decision stage. The study is restricted to ex-ante pre-screening before sukuk pricing, legal structuring, issuance approval and post-issuance impact reporting. The model therefore supports investment appraisal; it does not produce a credit rating, legal opinion or Shariah ruling.

Instrumental scope. The study focuses on sukuk because their asset-linked structure makes them analytically relevant to infrastructure appraisal. Conventional bonds, equity instruments, takaful products and general Islamic banking performance are excluded.

Methodological scope. The framework uses explainable multi-criteria decision-making rather than black-box machine learning. It does not automate fatwa issuance or replace human Shariah governance.

Empirical scope. The empirical demonstration is delimited to three to five representative scenarios and expert validation. This scale is appropriate for a master-level proposal whose contribution is framework development and feasibility testing, not full market generalization.

Jurisdictional scope. The study is designed as a transferable architecture, not as a universally binding regulatory model. Jurisdictional calibration is required because sukuk governance differs across Islamic finance markets.

14.2 Limitations of the Study

The study has five material limitations. None of them is hidden, and none invalidates the central contribution. They define how the results should be interpreted and where future research should extend the model.

Table 12 separates unavoidable limitations from mitigation strategies and shows how each weakness creates a defined future research pathway.

Table 12. Limitations, Mitigation Strategies and Future Research Pathways

Limitation	Risk for interpretation	Mitigation and future research path
Data availability	Proprietary CAPEX/OPEX, revenue and governance data may be unavailable.	Use public evidence first; label assumptions explicitly. Future: test with confidential issuer-level data.
Scenario size	Three to five scenarios limit transaction diversity.	Select cases for analytical variation. Future: replicate with larger sukuk transaction samples.
Expert sample	Weights may reflect the panel's jurisdictional and professional profile.	Use purposive sampling, consistency checks and sensitivity analysis. Future: compare expert panels across jurisdictions.
Maqasid and ESG proxies	Ethical and sustainability impact cannot be measured with perfect precision.	Use transparent proxies and expert validation. Future: validate indicators against post-issuance impact evidence.
MCDM sensitivity	Rankings may shift under alternative weights or aggregation rules.	Use MEREC/COPRAS as robustness checks and report divergence rather than suppressing it.

Note. The limitations are interpretation boundaries, not weaknesses hidden from the reader. Each limitation is paired with a mitigation strategy and a defined future research pathway.

14.3 Strategic Defense of the Limitations

The strongest limitation is data access. Smart infrastructure sukuk documentation may not disclose all project-level assumptions, especially revenue projections, operating expenditure, demand risk, contractual risk allocation and technical performance data. The study manages this limitation by using a transparent evidence hierarchy: public prospectuses and development-bank documents are preferred; missing inputs are replaced only by conservative calibrated assumptions; and synthetic inputs are labelled rather than presented as observed data.

The second limitation is expert-panel composition. Because the study requires Islamic finance, Shariah governance, ESG and infrastructure expertise, probability sampling is neither realistic nor methodologically appropriate. Purposive sampling is the correct design choice. Bias is mitigated through consistency screening, expert-category documentation and sensitivity analysis across alternative weight structures.

The third limitation concerns Maqasid measurement. Maqasid al-Shariah cannot be reduced to a mechanical score without interpretive risk. The study addresses this by treating Maqasid indicators as structured proxies, not theological rulings. The score supports evidence organization; it does not replace qualified Shariah deliberation.

The fourth limitation concerns generalizability. SC-SIIRS is not presented as a universal regulatory model. It is a transferable framework that requires jurisdictional calibration. This is a methodological strength because sukuk governance differs across markets and an honest framework must preserve room for regulatory variation.

14.4 Negative or Null Findings

The study remains useful if the empirical demonstration produces unstable rankings, weak expert consensus or divergence between TOPSIS and COPRAS outputs. Such outcomes would identify where Islamic sustainable finance lacks measurement consensus and where Shariah-governed decision support requires stronger data standards.

If no tested scenario qualifies as strongly investment-ready, the finding remains valuable. It would show that nominal eligibility for sustainable sukuk is not equivalent to project readiness, strengthening the case for disciplined pre-issuance appraisal rather than weakening the study contribution.

14.5 Future Research Pathway

The limitations of this study generate a clear research agenda. Future studies should extend the framework by:

- testing SC-SIIRS on larger samples of actual sukuk transactions across GCC, Malaysia, Indonesia and other Islamic finance jurisdictions;
- linking readiness scores to post-issuance outcomes such as impact reporting quality, distribution coverage, project completion and Shariah audit findings;
- developing jurisdiction-specific weighting schemes for different Shariah governance, disclosure and infrastructure policy environments;
- building an explainable AI layer only after the MCDM framework and governance controls have been validated; and
- examining whether SC-SIIRS can be adapted for other real-asset Islamic finance instruments beyond sukuk.

Overall, the study is deliberately bounded and methodologically honest. It does not promise universal prediction, theological finality or market-wide generalization. Its defensible contribution is

narrower and stronger: a transparent, project-level, Shariah-governed pre-screening framework whose limitations are visible, managed and convertible into future empirical research.

15. Research Timeline and Work Plan

This work plan is deliberately conservative. A purely six-month schedule would underestimate three high-risk stages of the study: ethics approval, expert recruitment and mixed-methods integration. The proposed twelve-month plan therefore protects the methodological core of the study while keeping the minimum viable scope realistic: three smart infrastructure cases, eight valid AHP expert responses and six expert interviews.

15.1 Planning Assumptions

- The timetable is designed for a master-level or early doctoral research window of twelve months.
- The core methodology remains fixed: conceptual framework finalization, ethics approval, pilot testing, secondary data collection, AHP weighting, TOPSIS ranking, MEREC/COPRAS robustness checks, semi-structured interviews, thematic analysis and final writing.
- The minimum viable empirical design is three project scenarios, eight AHP experts and six interviews. The aspirational target is five project scenarios, fifteen to twenty AHP experts and up to ten interviews.
- Buffers are placed before and after the highest-risk stages: ethics approval, recruitment, data access and integration of quantitative and qualitative findings.

15.2 Gantt-Ready Research Work Plan

Table 13 converts the study design into a twelve-month work plan organized around deliverables rather than vague activity labels.

Table 13. Twelve-Month Research Timeline

Phase / milestone	Timing	Core tasks	Deliverable or decision gate
Phase 1 - Scope lock and protocol refinement	Month 1	Finalize research scope, unit of analysis, minimum viable sample, case inclusion criteria and SC-SIIRS dimensions.	Approved research protocol and locked scope statement.
Phase 2 - Critical literature consolidation	Months 1-2	Complete critical literature review; map existing work on sukuk, Maqasid-ESG, smart infrastructure, MCDM and Shariah governance.	Critical matrix and finalized research literature

Phase / milestone	Timing	Core tasks	Deliverable or decision gate
Phase 3 - Ethics preparation and instrument design	Months 2-3	Prepare ethics application, participant information sheet, consent form, AHP questionnaire and interview guide.	Ethics submission package and draft research instruments.
Phase 4 - Ethics review buffer and pilot test	Months 3-4	Obtain ethics approval or exemption; pilot the AHP survey and interview guide with two to three reviewers; revise ambiguous criteria.	Ethics clearance or exemption; pilot-tested instruments.
Phase 5 - Secondary data and case construction	Months 4-5	Collect public sukuk, IsDB, World Bank, transport authority, ESG and PPP documents; construct three to five project scenarios.	Scenario database and evidence register.
Phase 6 - Expert recruitment and AHP survey	Months 5-6	Recruit Islamic finance, Shariah governance, ESG and infrastructure experts; collect pairwise comparison matrices.	Minimum eight valid AHP responses; aspirational fifteen to twenty.
Phase 7 - Quantitative modelling	Months 6-7	Calculate AHP weights, normalize project data, compute financial viability indicators and run TOPSIS ranking.	AHP weights, normalized decision matrix and primary project ranking.
Phase 8 - Robustness and sensitivity analysis	Months 7-8	Apply MEREC and COPRAS as robustness checks; vary key weights; document ranking reversals or stability.	Robustness report and sensitivity matrix.
Phase 9 - Qualitative interviews	Months 8-9	Conduct six to ten semi-structured interviews on framework validity, Shariah governance compatibility and institutional usability.	Interview transcripts or structured notes, anonymized and coded.
Phase 10 - Thematic analysis and integration	Months 9-10	Code interviews; integrate qualitative themes with quantitative scoring results; identify governance safeguards.	Integrated mixed-methods findings and revised SC-SIIRS framework.
Phase 11 - Writing and supervisor review	Months 10-11	Draft results, discussion, contribution, limitations and final methodology; submit full draft for supervisor review.	Complete thesis / proposal draft and revision list.
Phase 12 - Final	Month 12	Finalize references,	Final submission file,

Phase / milestone	Timing	Core tasks	Deliverable or decision gate
revision, formatting and defense preparation		figures, appendices, formatting, plagiarism check, submission package and oral defense materials.	defense slides and practitioner-ready summary.

Note. The schedule is structured around deliverables rather than activities alone. This makes progress auditable and preserves a hard final month for integration, formatting and defense preparation.

15.3 Milestones, Buffers and Scope-Control Gates

Table 14 defines the contingency gates that prevent delays from expanding the study beyond its feasible scope.

Table 14. Milestones and Contingency Gates

Milestone / risk point	Trigger	Corrective action
Ethics approval	No approval or exemption by the end of Month 4.	Continue desk-based case construction and literature consolidation; postpone primary data collection until clearance is obtained.
AHP expert recruitment	Fewer than eight valid expert responses by mid-Month 6.	Freeze the minimum viable panel if at least six high-quality experts respond; strengthen sensitivity analysis and disclose reduced panel size.
Real project data availability	Fewer than three sufficiently documented real cases by Month 5.	Use calibrated synthetic cases based on public prospectuses, IsDB/World Bank documents and transport authority reports.
Interview recruitment	Fewer than six completed interviews by Month 9.	Use a smaller qualitative validation sample and compensate with structured expert feedback from AHP comments and document analysis.
MCDM instability	TOPSIS and COPRAS produce materially different rankings.	Treat divergence as a robustness finding; report which criteria drive ranking sensitivity instead of forcing artificial convergence.
Writing compression	Analysis continues into Month 11.	Prioritize core thesis chapters; move extended scenario details, raw matrices and full coding tables to appendices.

Note. The contingency strategy protects feasibility by reducing empirical complexity when delays occur while preserving the main contribution: a transparent project-level SC-SIIRS decision-support framework.

15.4 Final Submission Logic

The final month is reserved for integration, formatting and defense preparation rather than new data collection. This is deliberate. A timeline that continues data collection until the submission deadline would be methodologically unsafe because it would leave no room for supervisor feedback, reference correction, figure verification, formatting, plagiarism checks or oral defense preparation.

The timeline therefore treats Month 12 as a hard closure period. By that stage, all primary analysis should be complete, all tables and figures should be finalized, and only editorial, formatting and defense tasks should remain.

Crash-test response. If ethics approval, expert recruitment and real project data all underperform simultaneously, the thesis remains viable by switching to the minimum design: desk-based framework validation, three calibrated scenarios, a reduced expert panel and expanded sensitivity analysis. The contribution would become a rigorously documented feasibility and framework-building study rather than a broad empirical validation exercise.

16. Conclusion

This proposal addresses a precise failure in Islamic sustainable infrastructure finance: projects can be Shariah-compliant in form, sustainability-labelled in language, or technologically attractive in design, while still remaining difficult to compare as sukuk-ready investment candidates. The missing step is a disciplined pre-structuring appraisal model that makes project-level trade-offs visible before legal, Shariah and capital-market resources are committed. SC-SIIRS responds to that gap by translating financial viability, Maqasid alignment, ESG contribution, smart infrastructure performance, Shariah governance and risk exposure into a transparent readiness architecture. Its sequential design - criteria operationalization, AHP-based expert weighting, TOPSIS project ranking, MEREC/COPRAS robustness checks and expert interviews - creates a direct route from research question to defensible output: an auditable decision-support framework that supports, rather than replaces, Shariah boards and investment committees.

The contribution is not another general claim that Islamic finance is compatible with sustainability. It is a move from compatibility to allocation discipline. Academically, the study shifts Islamic sustainable finance from institution-level and market-level discussion toward project-level appraisal. Methodologically, it shows how MCDM can be used in a Shariah-governed context without turning judgment into a black-box score. Practically, it gives Islamic banks, sukuk issuers, Shariah reviewers, regulators and infrastructure planners a structured way to distinguish projects that are merely eligible from projects that are documented, ethical, financially viable, technically credible, risk-aware and ready for further sukuk structuring. The final value of SC-SIIRS is that it turns Islamic sustainable infrastructure finance from a statement of principles into an auditable decision about which projects deserve to advance.

17. References

- AAOIFI & IFSB. (2022). Exposure draft: Revised Shari'ah governance framework for institutions offering Islamic financial services. <https://aaoifi.com/announcement/the-aaoifi-and-ifsb-jointly-issued-exposure-draft-of-ifsb-aaoifi-revised-shariah-governance-framework-for-institutions-offering-islamic-financial-services-rsgf/>
- AAOIFI. (2024, February 15). Deadline extension for industry feedback on exposure draft of Shari'ah Standard No. 62 "Sukuk". <https://aaoifi.com/announcement/deadline-extension-for-industry-feedback-on-exposure-draft-of-shariah-standard-no-62-sukuk/>
- Ahmed, H. (2011). Maqasid al-Shari'ah and Islamic financial products: A framework for assessment. *ISRA International Journal of Islamic Finance*, 3(1), 149-160. <https://journal.inceif.edu.my/index.php/ijif/article/view/116>
- American Psychological Association. (n.d.). Journal Article Reporting Standards (JARS). <https://apastyle.apa.org/jars>
- Beauchamp, T. L., & Childress, J. F. (2019). *Principles of biomedical ethics* (8th ed.). Oxford University Press.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77-101. <https://doi.org/10.1191/1478088706qp063oa>
- Chapra, M. U. (2008). *The Islamic vision of development in the light of Maqasid al-Shariah*. International Institute of Islamic Thought.
- Creswell, J. W., & Plano Clark, V. L. (2018). *Designing and conducting mixed methods research* (3rd ed.). SAGE.
- Diamond, D. W. (1984). Financial intermediation and delegated monitoring. *The Review of Economic Studies*, 51(3), 393-414. <https://doi.org/10.2307/2297430>
- Dusuki, A. W., & Abdullah, N. I. (2007). Maqasid al-Shari'ah, maslahah, and corporate social responsibility. *American Journal of Islam and Society*, 24(1), 25-45. <https://doi.org/10.35632/ajis.v24i1.415>
- EQUATOR Network. (n.d.). Reporting guidelines. <https://www.equator-network.org/reporting-guidelines/>

- European Commission. (2021, July 13). EU grants: How to complete your ethics self-assessment. https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/guidance/how-to-complete-your-ethics-self-assessment_en.pdf
- European Commission. (n.d.). Data protection explained. https://commission.europa.eu/law/law-topic/data-protection/data-protection-explained_en
- Freeman, R. E. (1984). Strategic management: A stakeholder approach. Pitman.
- Hwang, C.-L., & Yoon, K. (1981). Multiple attribute decision making: Methods and applications: A state-of-the-art survey. Springer. <https://doi.org/10.1007/978-3-642-48318-9>
- ICD & LSEG. (2025, October 15). LSEG and ICD announce findings of the 2025 Islamic Finance Development Indicator Report: Global Islamic finance assets projected to reach US\$9.7 trillion by 2029. <https://icd-ps.org/en/news/lseg-and-icd-announce-findings-of-the-2025-islamic-finance-development-indicator-report-global-islamic-finance-assets-projected-to-reach-us97-trillion-by-2029>
- ICMA, Islamic Development Bank, & LSEG. (2024). Guidance on green, social and sustainability sukuk. <https://www.icmagroup.org/assets/documents/Sustainable-finance/ICMA-IsDB-LSEG-Guidance-on-Green-Social-and-Sustainability-Sukuk-April-2024.pdf>
- ICMA. (2025). Green Bond Principles: Voluntary process guidelines for issuing green bonds. <https://www.icmagroup.org/assets/documents/Sustainable-finance/2025-updates/Green-Bond-Principles-GBP-June-2025.pdf>
- Islamic Development Bank. (2025). Sustainable finance framework. <https://www.isdb.org/publications/isdb-sustainable-finance-framework-july-2025>
- Islamic Financial Services Board. (2025). Islamic financial services industry stability report 2025. <https://www.ifsb.org/wp-content/uploads/2025/05/IFSI-Stability-Report-2025.pdf>
- Keen, P. G. W., & Scott Morton, M. S. (1978). Decision support systems: An organizational perspective. Addison-Wesley.
- Keshavarz-Ghorabae, M., Amiri, M., Zavadskas, E. K., Turskis, Z., & Antucheviciene, J. (2021). Determination of objective weights using a new method based on the removal

effects of criteria (MEREC). *Symmetry*, 13(4), Article 525.
<https://doi.org/10.3390/sym13040525>

National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. (1979). *The Belmont Report: Ethical principles and guidelines for the protection of human subjects of research*. U.S. Department of Health and Human Services. <https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html>

Papa, C., & Rossi, N. (2022). Smart cities and sustainable finance: The Islamic perspective. *European Journal of Islamic Finance*, 9(2), 18-26. <https://doi.org/10.13135/2421-2172/6167>

Ramli, N. A., Mat Nor, F., Ab. Aziz, M. R., Shahwan, S., Marzuki, A., Ali Basah, M. Y., & Sabri, H. (2024). Application of Maqasid al-Shariah in responsible investment (MSRI) and practices towards ESG. *Journal of Infrastructure, Policy and Development*, 8(16), Article 5864. <https://doi.org/10.24294/jipd5864>

Ross, P. T., & Bibler Zaidi, N. L. (2019). Limited by our limitations. *Perspectives on Medical Education*, 8(4), 261-264. <https://doi.org/10.1007/s40037-019-00530-x>

Saaty, R. W. (1987). The analytic hierarchy process-what it is and how it is used. *Mathematical Modelling*, 9(3-5), 161-176. [https://doi.org/10.1016/0270-0255\(87\)90473-8](https://doi.org/10.1016/0270-0255(87)90473-8)

Schoenmaker, D., & Schramade, W. (2019). *Principles of sustainable finance*. Oxford University Press.

Science and Technology Facilities Council. (2021). *Guidelines for projects peer review proposal*. UK Research and Innovation. <https://www.ukri.org/wp-content/uploads/2021/07/STFC-100921-GuidelinesforProjectsPeerReviewProposal-V2.pdf>

Simon, H. A. (1960). *The new science of management decision*. Harper & Brothers.

Theofanidis, D., & Fountouki, A. (2018). Limitations and delimitations in the research process. *Perioperative Nursing*, 7(3), 155-163. <https://doi.org/10.5281/zenodo.2552022>

U.S. Department of Health and Human Services, Office for Human Research Protections. (n.d.). *Informed Consent FAQs*. <https://www.hhs.gov/ohrp/regulations-and-policy/guidance/faq/informed-consent/index.html>

- World Bank & Islamic Development Bank. (2019). Reference guide: Islamic finance for infrastructure PPP projects. World Bank Group. <https://ppp.worldbank.org/library/reference-guide-islamic-finance-infrastructure-ppp-projects>
- World Bank & Islamic Development Bank. (2025). Islamic finance and climate agenda: From green sukuk to climate-resilient development. World Bank. <https://openknowledge.worldbank.org/entities/publication/25a635f9-254e-4564-9fc7-e895b4918504>
- Ye, F., Chen, Y., Li, L., Li, Y., & Yin, Y. (2022). Multi-criteria decision-making models for smart city ranking: Evidence from the Pearl River Delta region, China. *Cities*, 128, Article 103793. <https://doi.org/10.1016/j.cities.2022.103793>
- Zain, F. A. M., Muhamad, S. F., Abdullah, H., Sheikh Ahmad Tajuddin, S. A. F., & Wan Abdullah, W. A. (2024). Integrating environmental, social and governance (ESG) principles with Maqasid al-Shariah: A blueprint for sustainable takaful operations. *International Journal of Islamic and Middle Eastern Finance and Management*, 17(3), 461-484. <https://doi.org/10.1108/IMEFM-11-2023-0422>
- Zavadskas, E. K., Kaklauskas, A., & Sarka, V. (1994). The new method of multicriteria complex proportional assessment of projects. *Technological and Economic Development of Economy*, 1(3), 131-139.

Appendix A. SC-SIIRS Criteria and Indicator Matrix

This appendix translates the conceptual structure of SC-SIIRS into operational criteria. Its role is practical: it shows how each dimension can be observed, scored and checked before being used in AHP weighting, TOPSIS ranking and robustness analysis. The matrix also clarifies whether a criterion is treated as a benefit, cost, penalty or minimum gate.

Table A1. SC-SIIRS Criteria and Indicator Matrix

Dimension	Indicator examples	Measurement logic	Direction
Shariah compliance and governance	Asset linkage; revenue permissibility; ownership or usufruct logic; Shariah review trail	Assessed through documentary evidence and expert review of permissibility, auditability and governance traceability	Gate / benefit
	Financial viability	CAPEX; OPEX; cash-flow proxy; payback period; NPV/IRR where available; distribution coverage logic	Scored through quantitative project data, normalized financial indicators and scenario-level financial assumptions
Maqasid al-Shariah impact	Public welfare; harm reduction; accessibility; safety; preservation of wealth, life and public interest	Evaluated through structured expert scoring supported by project-level evidence and policy justification	Benefit
	ESG contribution	Emissions reduction; resource efficiency; social inclusion; disclosure quality; governance transparency	Assessed through ESG reports, impact proxies, sustainability documentation and disclosure quality
Smart infrastructure performance	Reliability; interoperability; digital access; resilience; mobility or utility performance gains	Measured through technical KPIs, operational benchmarks and scenario-level infrastructure evidence	Benefit
	Risk exposure	Regulatory risk; demand risk; implementation risk; technology risk; reputational risk; Shariah ambiguity	Scored as a risk penalty based on documented uncertainty, expert assessment and scenario-level risk flags

Note. SC-SIIRS refers to the Shariah-Compliant Smart Infrastructure Investment Readiness Score. The model is applied only after minimum Shariah and documentation gates have been satisfied. The resulting score should be interpreted as a pre-screening and decision-support instrument. It can indicate whether a project is ready for deeper review, but it does not replace Shariah board approval, legal due diligence or investment committee judgment.

Appendix B. Secondary Data Extraction Template for Project Scenarios

To avoid selective evidence collection, each project scenario is documented through the same extraction grid. This makes comparisons more transparent and reduces the risk of giving stronger scores to projects simply because more information is available for them. The template also forces synthetic or calibrated assumptions to be identified before scoring begins.

Table B1. Secondary Data Extraction Template for Project Scenarios

Extraction field	What must be recorded	Evidence/source requirement
Scenario identification	Project name, sector, location, infrastructure type, issuer or sponsor if public	Prospectus, development-bank report, public authority document or labelled synthetic case
Asset and sukuk logic	Underlying asset, usufruct/service logic, revenue source and potential sukuk compatibility	Sukuk documents, Shariah certificate, legal summary or documented assumption
Financial inputs	CAPEX, OPEX, cash-flow proxy, payback, NPV/IRR where available, coverage logic	Financial model, project report, feasibility study or calibrated synthetic values
Maqasid evidence	Public welfare, access, safety, harm reduction, inclusion and preservation objectives	Policy document, project impact note, expert scoring justification
ESG evidence	Environmental, social and governance contribution plus disclosure quality	ESG report, impact report, sustainability framework or sector benchmark
Smart infrastructure KPIs	Reliability, interoperability, digital access, resilience and performance improvement	Technical report, transport/utility authority data or operational benchmark
Risk flags	Regulatory, implementation, demand, technology, reputational and Shariah risk	Risk register, expert assessment, public disclosure or documented scenario assumption
Data quality	Completeness, source reliability, date, uncertainty and missing fields	Source log and researcher assessment

Note. When real project data are incomplete, the scenario must be explicitly labelled as synthetic, calibrated or partially observed. No synthetic value should be presented as observed market data.

Appendix C. AHP Expert Pairwise Comparison Questionnaire

This appendix presents the AHP questionnaire used to derive the relative weights of the SC-SIIRS dimensions. Because the model combines sukuk structuring, Shariah governance, infrastructure finance, ESG assessment and MCDM-based decision support, the questionnaire is intended for a specialized expert panel rather than a general-public sample. The objective is not to measure opinion, but to formalize professional judgment in a transparent and consistency-tested way.

C.1 Instructions to Expert Respondents

Please compare the importance of each criterion pair for pre-screening smart infrastructure projects before Shariah-compliant sukuk structuring.

1. Use the 1–9 AHP scale shown in Table C1.
2. A value of 1 means that both criteria are equally important.
3. A value of 9 means that one criterion is extremely more important than the other.
4. Base your answers on your professional expertise, not on the expected preference of a specific institution.
5. Responses with severe inconsistency may be reviewed, clarified or excluded according to the consistency-screening rule defined in the methodology.
6. Estimated completion time: 15–20 minutes.

Table C1. AHP Pairwise-Comparison Scale

Scale value	Meaning	Use in this study
1	Equal importance	Both criteria contribute equally to sukuk readiness
3	Moderate importance	One criterion is moderately more important
5	Strong importance	One criterion is strongly more important
7	Very strong importance	One criterion clearly dominates the other
9	Extreme importance	One criterion is overwhelmingly more important
2, 4, 6, 8	Intermediate values	Used when judgment falls between adjacent levels
Reciprocal	Opposite preference	Used when the column criterion is preferred to the row criterion

Note. AHP refers to the Analytic Hierarchy Process. The scale converts expert pairwise judgments into criterion weights. Responses are screened for consistency before aggregation.

C.2 Dimension-Level Pairwise Comparison Matrix

The matrix below is used to compare the six SC-SIIRS dimensions. Experts complete only one side of the matrix. Reciprocal values are calculated during data processing.

Table C2. Dimension-Level AHP Pairwise-Comparison Matrix

	Criteria	Sharia	Financ	Maqas	ESG	Smart	Risk
a	h governance	ial viability	id impact	contribution	infrastructure performance	exposure	
	Sharia	1					
	h governance						
	Financ		1				
	ial viability						
	Maqas			1			
	id impact						
	ESG				1		
	contribution						
	Smart					1	
	infrastructure performance						
	Risk						1
	exposure						

Note. Blank cells are completed by expert respondents. Reciprocal values are calculated during the AHP data-processing stage.

C.3 Open Validation Questions

The questionnaire includes open questions to capture issues that cannot be reduced to numerical comparisons. These questions help identify missing dimensions, vague indicators and practical adoption barriers.

1. Are any SC-SIIRS dimensions redundant, missing or wrongly defined?
2. Which criteria should be treated as non-compensatory gates rather than weighted dimensions?
3. Which indicators are most difficult to score reliably before sukuk issuance?
4. What explanation format would make the final score acceptable for Shariah governance review?

Note. The questionnaire should be pilot-tested before full deployment to check clarity, completion time and respondent burden.

Appendix D. Semi-Structured Expert Interview Guide

This interview guide is used to validate the usability, explainability and Shariah-governance compatibility of SC-SIIRS after the quantitative scoring architecture has been specified. The interviews are semi-structured: the same core themes are covered across participants, but question order may be adapted to the respondent's expertise.

Table D1. Semi-Structured Expert Interview Guide

Theme	Core question	Optional probes
Model relevance	Which SC-SIIRS dimensions are essential for assessing smart infrastructure sukuk readiness?	Are any dimensions over-weighted, under-weighted or missing?
Maqasid validity	Which Maqasid indicators are defensible at project level?	Which indicators risk becoming vague or symbolic?
Shariah governance	How should a Shariah board treat an MCDM readiness score?	Advisory input, formal criterion, red-flag tool or decision-support evidence?
Data availability	What data would Islamic banks, issuers or infrastructure authorities realistically provide before issuance?	Which indicators are likely to suffer from missing or low-quality data?
Greenwashing and Shariah-washing	What risks could make a project appear eligible while remaining weak in substance?	What evidence should be required to reduce this risk?
Explainability	What explanation format would make the score understandable to non-technical governance actors?	Dashboard, audit trail, criterion-by-criterion narrative or traffic-light band?
Adoption conditions	What institutional conditions are necessary before SC-SIIRS can be used responsibly?	Governance owner, review process, documentation burden and update cycle.

Note. The guide is designed to support comparability without forcing identical conversations. Each interview should cover all core themes, while allowing experts to expand on the areas where they have the strongest professional knowledge.

Appendix E. Participant Information Sheet

Title of study: A Shariah-Governed MCDM Readiness Framework for Smart Infrastructure Sukuk.

Researcher: El Ghali Sany. Supervisor :

Purpose of the study

This study develops and validates SC-SIIRS, a project-level decision-support framework for pre-screening smart infrastructure projects before Shariah-compliant sukuk structuring. The framework combines Shariah governance, financial viability, Maqasid al-Shariah, ESG contribution, smart infrastructure performance and risk exposure.

The study does not issue investment advice, legal advice or Shariah certification. Its purpose is academic: to examine whether a structured MCDM framework can improve transparency and comparability in the early assessment of smart infrastructure sukuk projects.

Why you are invited

You are invited because of your professional or academic expertise in one or more of the following areas:

- Islamic finance
- Sukuk structuring
- Shariah governance
- Sustainable finance
- ESG assessment
- Infrastructure finance
- Smart infrastructure
- MCDM-based decision support

Your expertise is relevant because the study relies on professional judgment to test whether the proposed criteria are meaningful, usable and defensible in practice.

What participation involves

Participation may involve one or both of the following:

1. completing an AHP pairwise-comparison questionnaire;
2. taking part in a semi-structured expert interview.

The expected time commitment is approximately:

- 15–20 minutes for the questionnaire;
- 30–45 minutes for the interview.

The interview may be audio-recorded only with your consent. If you do not consent to recording, written notes may be used instead.

Voluntary participation and withdrawal

Participation is voluntary. You may refuse to answer any question and may withdraw from the study without penalty.

Withdrawal of identifiable interview data is possible until transcripts have been anonymized and integrated into aggregated analysis. After anonymization and aggregation, it may no longer be possible to remove individual contributions because they will no longer be directly attributable to a specific participant.

Risks and benefits

The study involves minimal risk. The main risks are professional sensitivity and deductive identification, especially because the expert community in Islamic finance and sukuk structuring may be relatively small.

The potential benefit is contribution to a more transparent appraisal framework for Islamic sustainable infrastructure finance.

Confidentiality

Direct identifiers will be removed from transcripts, questionnaires and reports. Participants will be referred to using codes such as:

- Expert 01
- Expert 02
- Expert 03

Institutional affiliations will be generalized unless explicit permission is given to identify them.

Appendix F. Informed Consent Form

Study title: A Shariah-Governed MCDM Readiness Framework for Smart Infrastructure Sukuk.

Participant code: _____ Date: _____

Table F1. Informed Consent Checklist for Expert Participants

Consent statement	Yes / No / Initials
I confirm that I have read and understood the participant information sheet.	
I have had the opportunity to ask questions about the study.	
I understand that my participation is voluntary and that I may withdraw without penalty.	
I understand the type of data that will be collected through the questionnaire and/or interview.	
I consent to my anonymized responses being used for academic analysis and reporting.	
I understand that direct identifiers will be removed and that quotations may be paraphrased or anonymized to reduce deductive identification risk.	
I consent to audio recording of the interview. If no, written notes may be used instead.	
I understand that SC-SIIRS is a research decision-support framework and not investment advice, legal advice or Shariah certification.	

Participant name / code: _____

Signature or electronic confirmation: _____

Researcher signature: _____

Appendix G. Data Management, Anonymization and Storage Protocol

This protocol explains how participant data and project-scenario data will be handled during the study. Its purpose is to protect confidentiality, preserve auditability and reduce the risk that model outputs are misused or interpreted beyond their intended academic scope.

Table G1. Data Management, Anonymization and Storage Protocol

Process area	Procedure	Control objective
Data minimization	Collect only data needed for recruitment, consent, expert weighting, exposure interview analysis and auditability.	Avoid unnecessary personal-data
Participant coding	Replace participant names with codes such as Expert 01, Expert 02 and Expert 03.	Reduce direct identification risk
Separation of files	Store consent forms and contact details separately from questionnaires, identity and research responses transcripts and analysis files.	Prevent direct linkage between
Audio recordings	Use recordings only where consent is granted; delete after transcription and verification according to institutional policy.	Limit retention of identifiable voice data
Anonymized transcripts	Remove names, institutions, confidential project references and highly identifying job descriptions.	Control deductive disclosure
AHP data	Store pairwise-comparison matrices under participant codes only.	Preserve auditability while protecting participant identity
Storage location	Use encrypted, password-protected and university-approved storage access wherever available.	Protect data against unauthorized
Access control	Restrict identifiable data to the researcher and supervisor or approved research team members.	Limit internal exposure
Retention and destruction	Retain anonymized analysis files according to institutional policy; delete identifiable files after the approved retention period.	Maintain compliance and avoid indefinite storage
Breach response	Secure the affected file or device, change credentials, notify the supervisor or institutional contact and follow applicable reporting procedures.	Contain and report potential data exposure

Note. This protocol should be updated if the host institution imposes a specific retention period, storage platform, data-protection procedure or consent-template requirement.

Appendix H. Ethics Approval Status and Self-Assessment Statement

This appendix records the ethical status of the project without implying that approval has already been granted. It also identifies the actions required before any expert questionnaire, interview or primary data collection activity begins.

Table H1. Ethics Approval Status and Self-Assessment Checklist

Ethics item	Current status	Required action before data collection
Human participants only	Yes — adult expert participants	Submit the questionnaire, interview guide, participant information sheet and consent form for review
Vulnerable participants	No vulnerable population targeted	Confirm exclusion criteria in the ethics application
Primary data collection	Not started	Do not begin recruitment, AHP survey or interviews before approval or exemption
Informed consent	Draft included in Appendix F	Adapt to institutional template and obtain signed or electronic consent
Participant information	Draft included in Appendix E	Provide to participants before consent
Confidentiality and anonymization	Protocol included in Appendix G	Apply participant coding and remove direct identifiers
Ethics approval letter	Pending / not yet attached	Insert the official approval letter once obtained, or attach formal exemption confirmation

Self-assessment statement

The proposed study is low-risk because it involves adult expert participants, professional judgment, non-clinical interviews and project-level decision-support analysis. The main ethical risks are confidentiality, deductive identification and interpretive misuse of SC-SIIRS.

These risks are addressed through informed consent, participant coding, data minimization, separate storage of identifying information, anonymized reporting and explicit limitation of the model as an advisory research framework.

Approval condition

Primary data collection will begin only after the relevant institutional ethics committee or research review authority has approved the study or confirmed that it is exempt from full review.